Lecture 15 (10/19)

More examples of multiple predictor regression

Last Week:

- In observational data (unlike a randomized trial), other factors can affect both X and Y. These are called confounding variables.
- Multiple-predictor regression $Y = \beta_0 + \sum_k \beta_k X_k + \epsilon$: β_k is the difference in average Y per unit of X_k controlling for (meaning holding fixed) the other predictors.
- Omitted variable bias is the difference between the single-predictor slope β_1' and the multiple-regression slope β_1. We also saw adding a predictor never decreases ρ^2 (or R^2).
- Examples: test scores and mother info, violent crime, and baseball (the second two were examples of Simpson's paradox).

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Today:

1) Omitted variables in AR processes
2) Incumbency advantage in congressional elections; categorical predictors
3) Leaving out all predictors but one
4) Leaving out one predictor

Recall the AR(1) process

$$Y_t = \rho Y_{t-1} + \epsilon_t.$$

Assume the process is stationary (all Y_t have the same marginal distribution). Suppose data is generated from an AR(1) process, fit to the model

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \epsilon'_t.$$

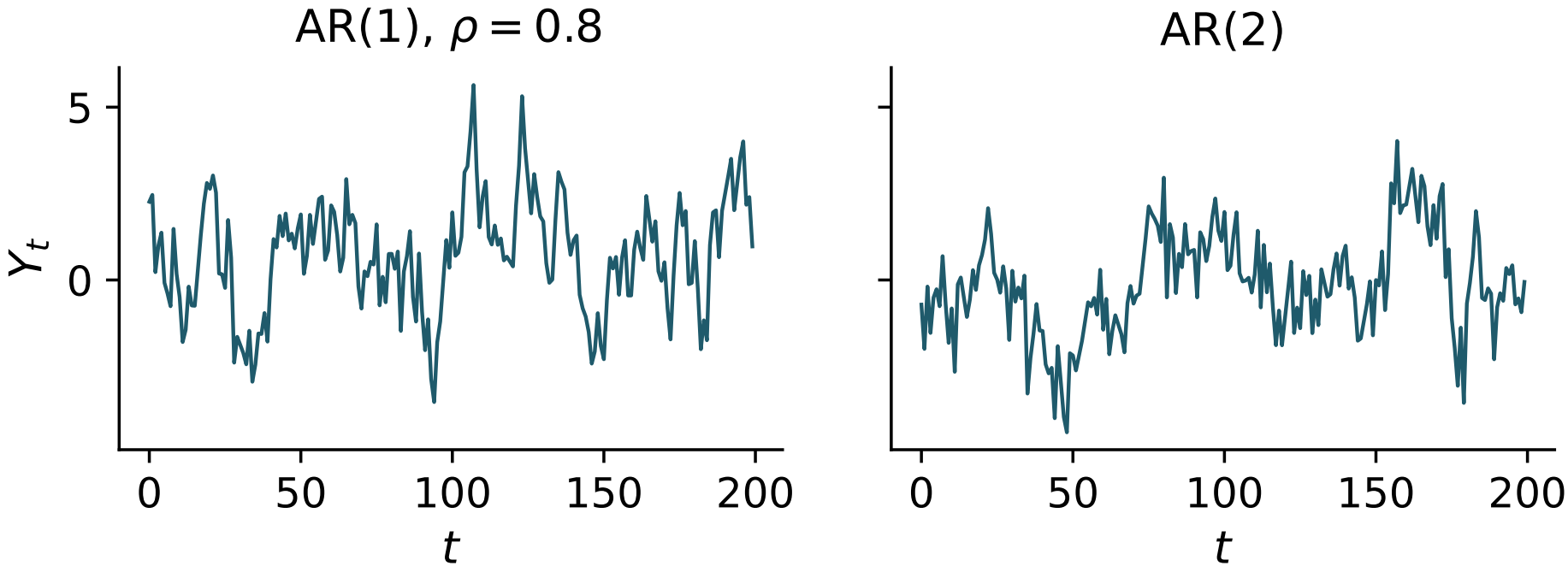
Question

Assuming a large amount of data, what will $\hat{\beta}_1$ and $\hat{\beta}_2$ be? Derive the regression coefficient when Y_{t-1} is omitted from the model using OVB. Then generalize this to obtain the regression coefficient between Y_{t-k} and Y_t.

An AR(2) process

Now let Y_t depend on *two* previous values:

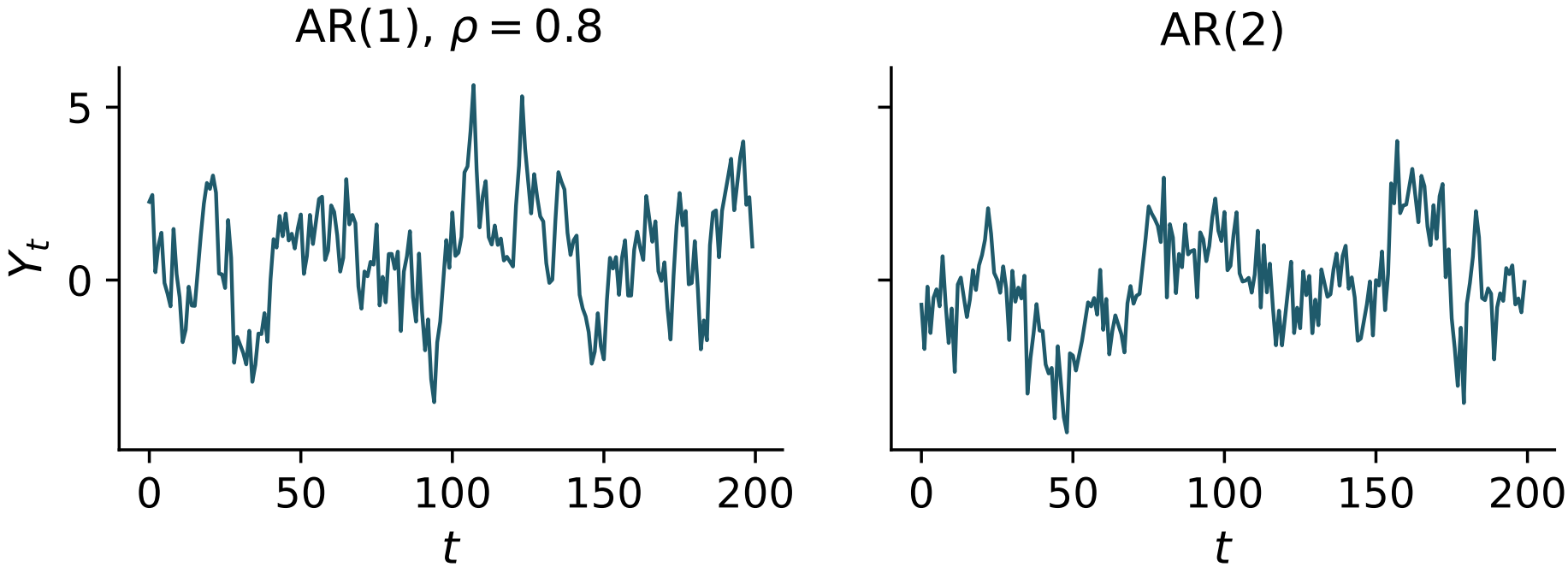
$$Y_t = 0.5\, Y_{t-1} + 0.3\, Y_{t-2} + \epsilon_t, \qquad \epsilon_t \sim \text{Normal}(0, 1).$$



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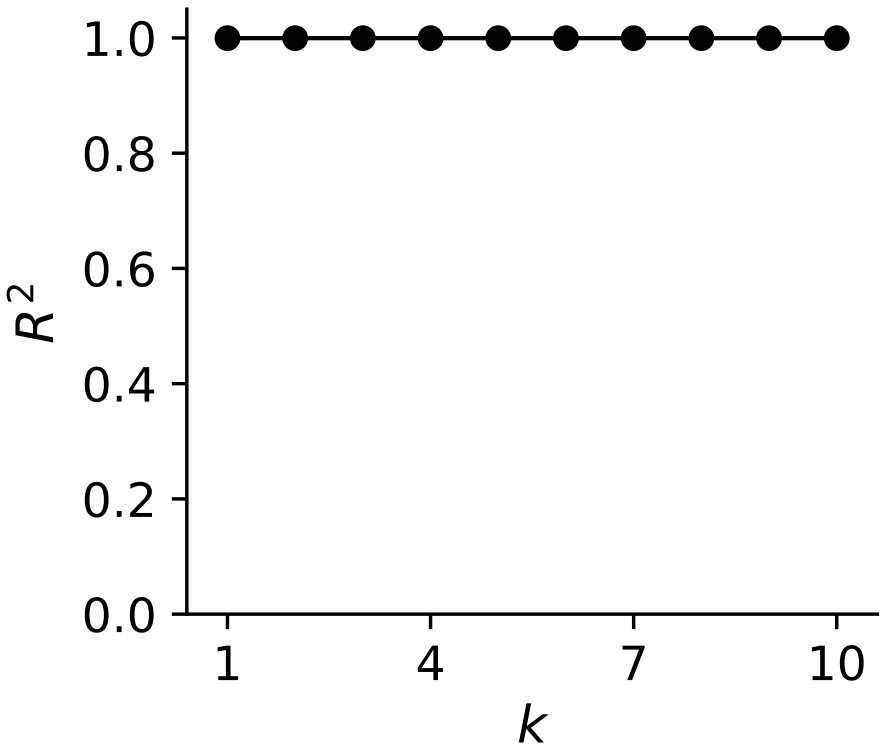


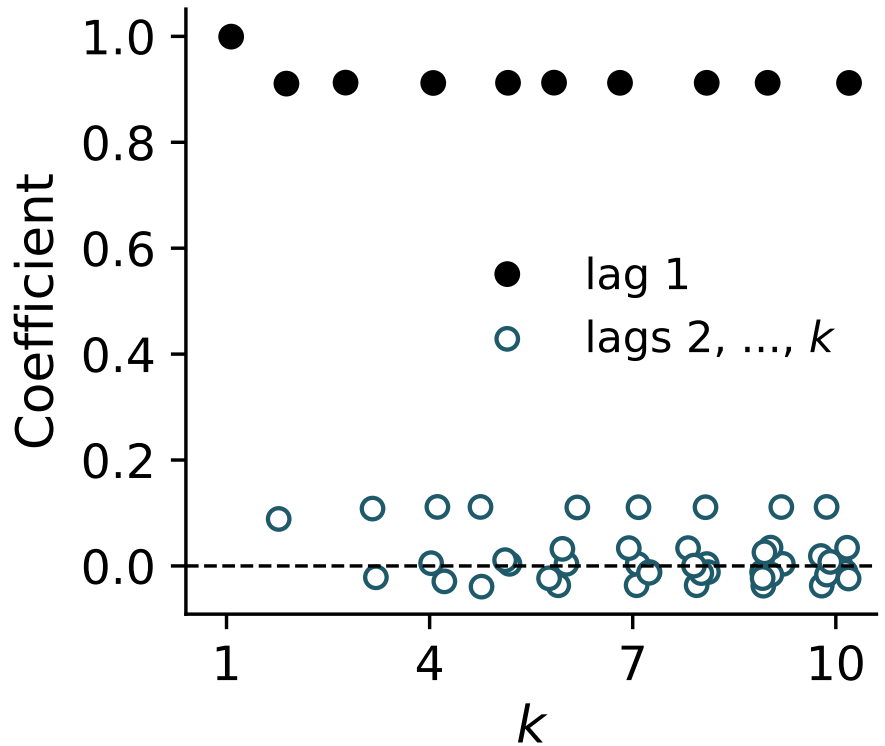
Question

What is the slope of the single-predictor regression of Y_t on Y_{t-1} (with Y_{t-2} excluded)?

Bitcoin: AR(k) models

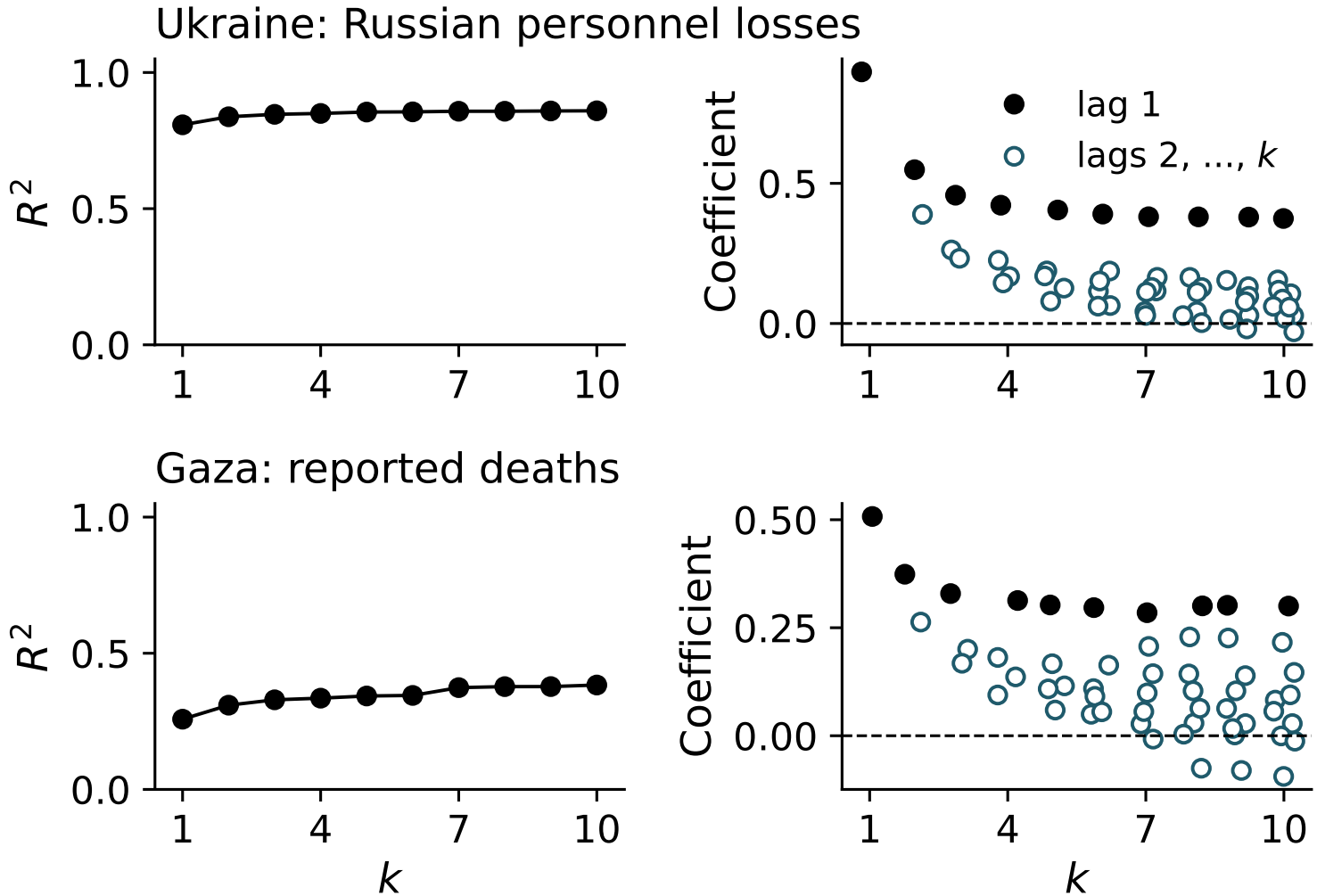
$Y_t = \log_{10}$(daily bitcoin price), regressed on its previous k values $Y_{t-1}, \ldots, Y_{t-k}$ for $k = 1, \ldots, 10$ (the same 4283 days in every fit):





Casualty counts: AR(k) models

Daily counts (changes in the cumulative series), regressed on the previous k days, $k = 1, \ldots, 10$:



Example

How big is the incumbency advantage? US House elections, 435 districts (from *Regression and Other Stories*):

- Y (`vote`): Democratic share of the two-party vote in 1988;
- X_{past} (`past_vote`): Democratic share in 1986;
- X_{inc} (`inc`): $+1$ if a Democrat is the incumbent, -1 if a Republican is, 0 for an open seat.

Uncontested races are set to 0.75 or 0.25.

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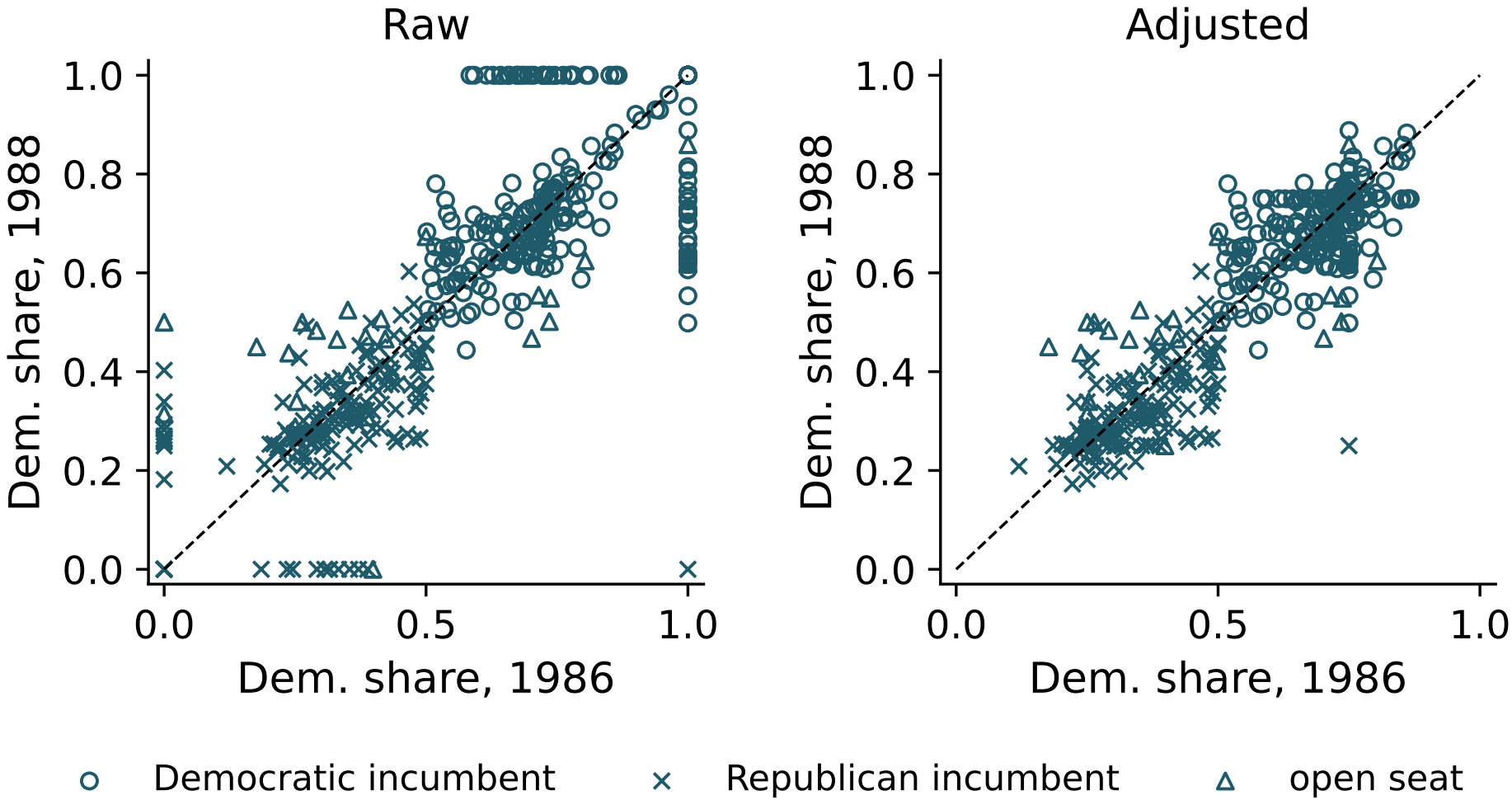
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Question

What is at least one limitation of this model? (You can assume this is the only data we have, so the limitation is about the assumptions)

1988 vs. 1986 vote



Each point is one district. The dashed line is $y = x$. Adjusted: uncontested races set to 0.25 or 0.75.

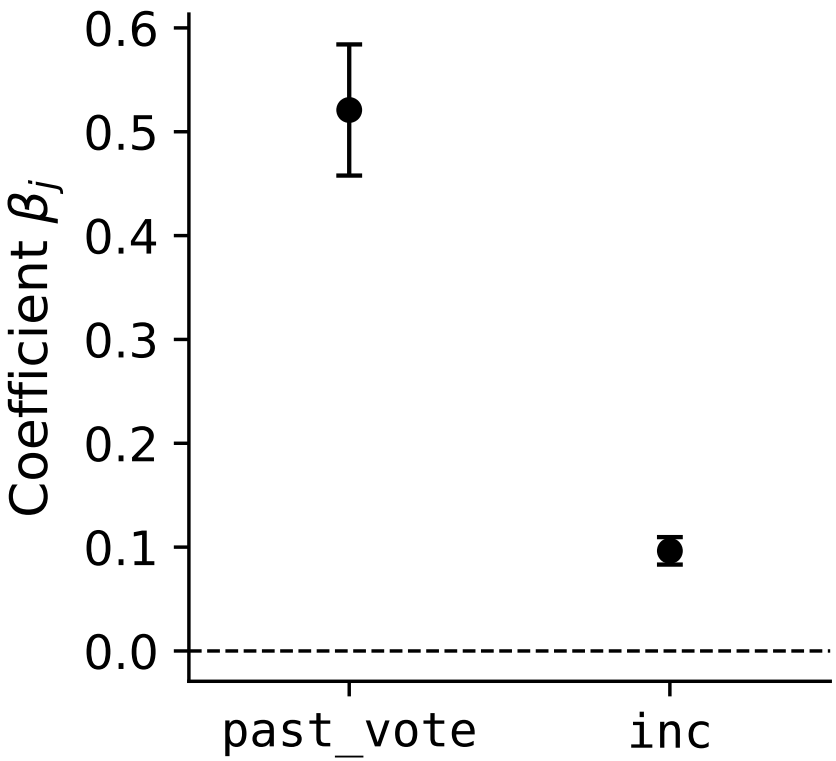
Dep. Variable:			vote	R-squared:		0.819
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.5063	0.004	121.695	0.000	0.498	0.514
inc	0.1906	0.004	44.210	0.000	0.182	0.199
Dep. Variable:			vote	R-squared:		0.887
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.2380	0.017	14.068	0.000	0.205	0.271
past_vote	0.5210	0.032	16.169	0.000	0.458	0.584
inc	0.0964	0.007	14.284	0.000	0.083	0.110

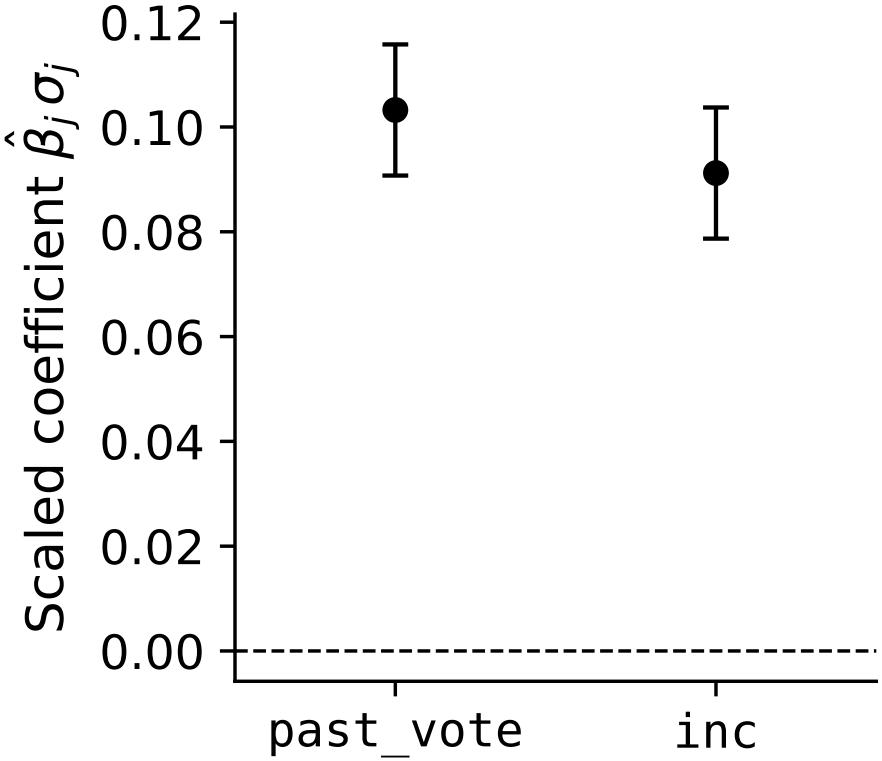
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```

Question

Interpret the coefficient on `inc` in each model. Approximately what is the regression coefficient of `inc` when `past_vote` is the response variable? Compare the importance of the two predictors. The standard deviations of the predictors are $\sigma_{\text{past}} \approx 0.20$ and $\sigma_{\text{inc}} \approx 0.95$.

Coefficients vs. scaled coefficients





Categorical predictors

- `inc` is categorical, which means it takes values in a set rather than on a linear scale. That is, changing from a Republican incumbent to an open seat may not have the same effect on Y as changing from an open seat to a Democratic incumbent.

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  - X_{dem} (`dem_inc`): Democratic incumbent;
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$$Y = \beta_0 + \beta_{\text{past}} X_{\text{past}} + \beta_{\text{dem}} X_{\text{dem}} + \beta_{\text{rep}} X_{\text{rep}} + \beta_{\text{open}} X_{\text{open}} + \epsilon$$

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The model above has a problem, which is seen when we try to answer the following question.

Question

What is the interpretation of β_{dem}?

Incumbency with dummy variables

Solution: leave out one category, the baseline (here, open seats). We compare the linear and categorical models:

$$Y = \beta_0 + \beta_{\text{past}} X_{\text{past}} + \beta_{\text{inc}} X_{\text{inc}} + \epsilon,$$

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past_vote	0.5270	0.032	16.260	0.000	0.463	0.591
dem_inc	0.0760	0.015	5.089	0.000	0.047	0.105
rep_inc	-0.1152	0.014	-8.227	0.000	-0.143	-0.088

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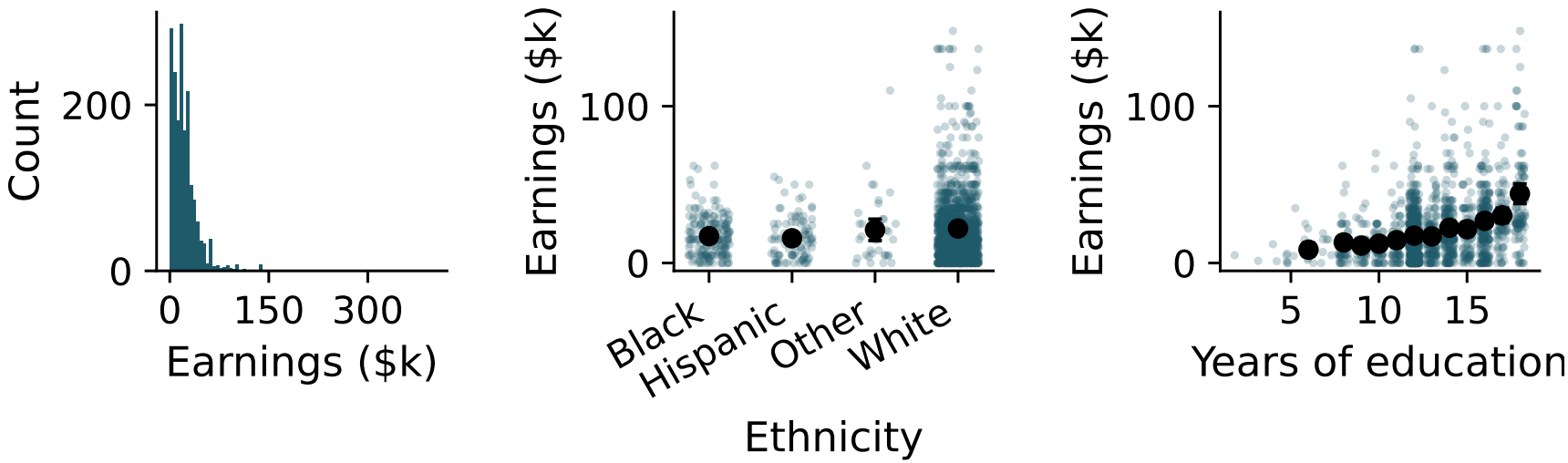
Which choice of β_{dem} and β_{rep} gives back the linear model? Compare $\hat{\beta}_{\text{dem}}$ and $\hat{\beta}_{\text{rep}}$ with $\hat{\beta}_{\text{inc}}$. Is the incumbency advantage the same for both parties?

Example

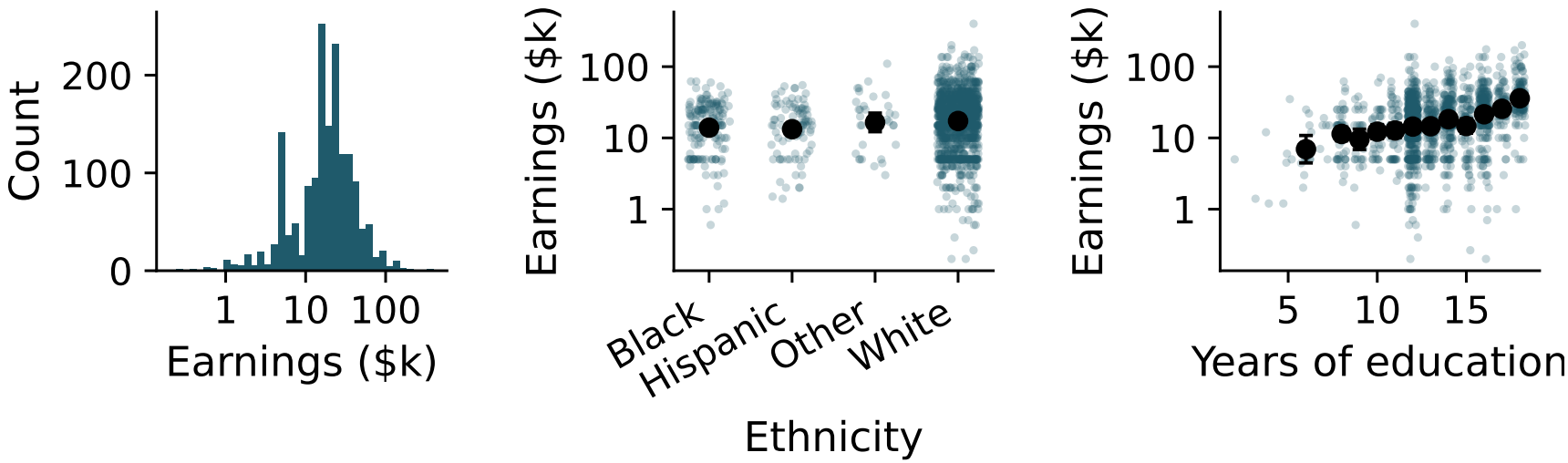
How are earnings related to gender and education, and what else could explain these associations? Survey of 1814 US adults (from *Regression and Other Stories*):

- Y (`earnk`): earnings in \$1000s;
- $X_{\text{hispanic}}, X_{\text{other}}, X_{\text{white}}$ (`hispanic`, `other`, `white`): dummy variables for ethnicity, with Black as the baseline;
- X_{male} (`male`), X_{height} (`height`, inches), X_{educ} (`education`, years), X_{age} (`age`, years).

Earnings vs. ethnicity and education



Linear scale. Note: 4 respondents earning more than $160k are not shown.



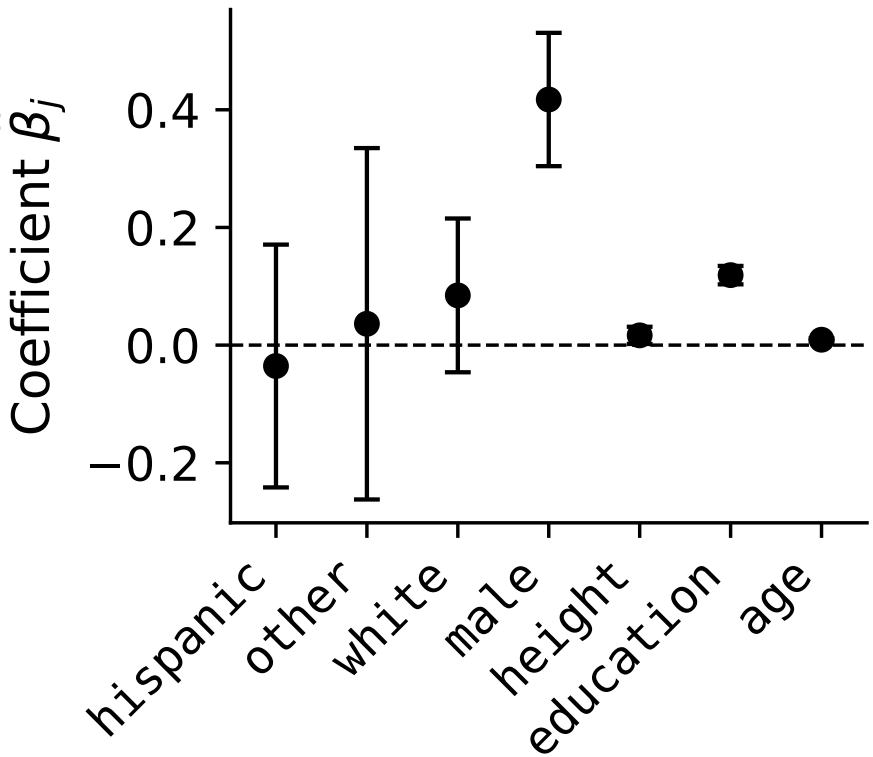
Log scale using 1627 respondents with positive earnings. Black dots: averages with 95% CIs.

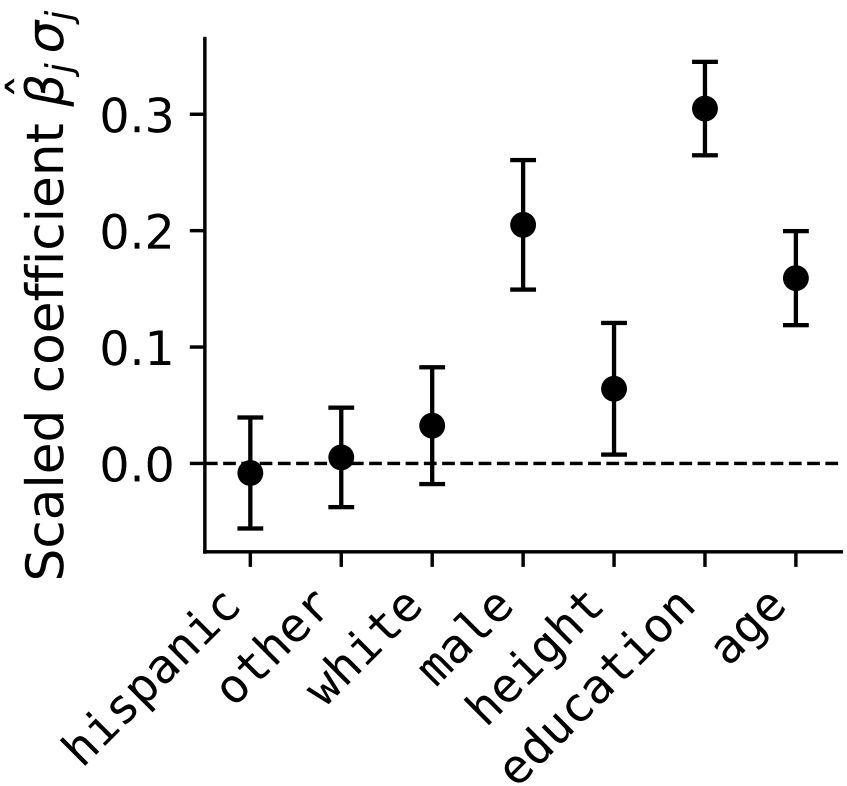
Earnings with all predictors

$$\log(\texttt{earnk}) = \beta_0 + \beta_{\text{hispanic}} X_{\text{hispanic}} + \beta_{\text{other}} X_{\text{other}} + \beta_{\text{white}} X_{\text{white}} + \beta_{\text{male}} X_{\text{male}} + \beta_{\text{height}} X_{\text{height}} + \beta_{\text{educ}} X_{\text{educ}} + \beta_{\text{age}} X_{\text{age}} + \epsilon$$

Dep. Variable:		log_earnk	R-squared:			0.211
	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.5156	0.496	-1.039	0.299	-1.489	0.458
hispanic	-0.0355	0.105	-0.337	0.736	-0.242	0.171
other	0.0363	0.152	0.238	0.812	-0.263	0.335
white	0.0845	0.067	1.268	0.205	-0.046	0.215
male	0.4175	0.058	7.222	0.000	0.304	0.531
height	0.0166	0.007	2.223	0.026	0.002	0.031
education	0.1189	0.008	14.885	0.000	0.103	0.135
age	0.0092	0.001	7.724	0.000	0.007	0.012

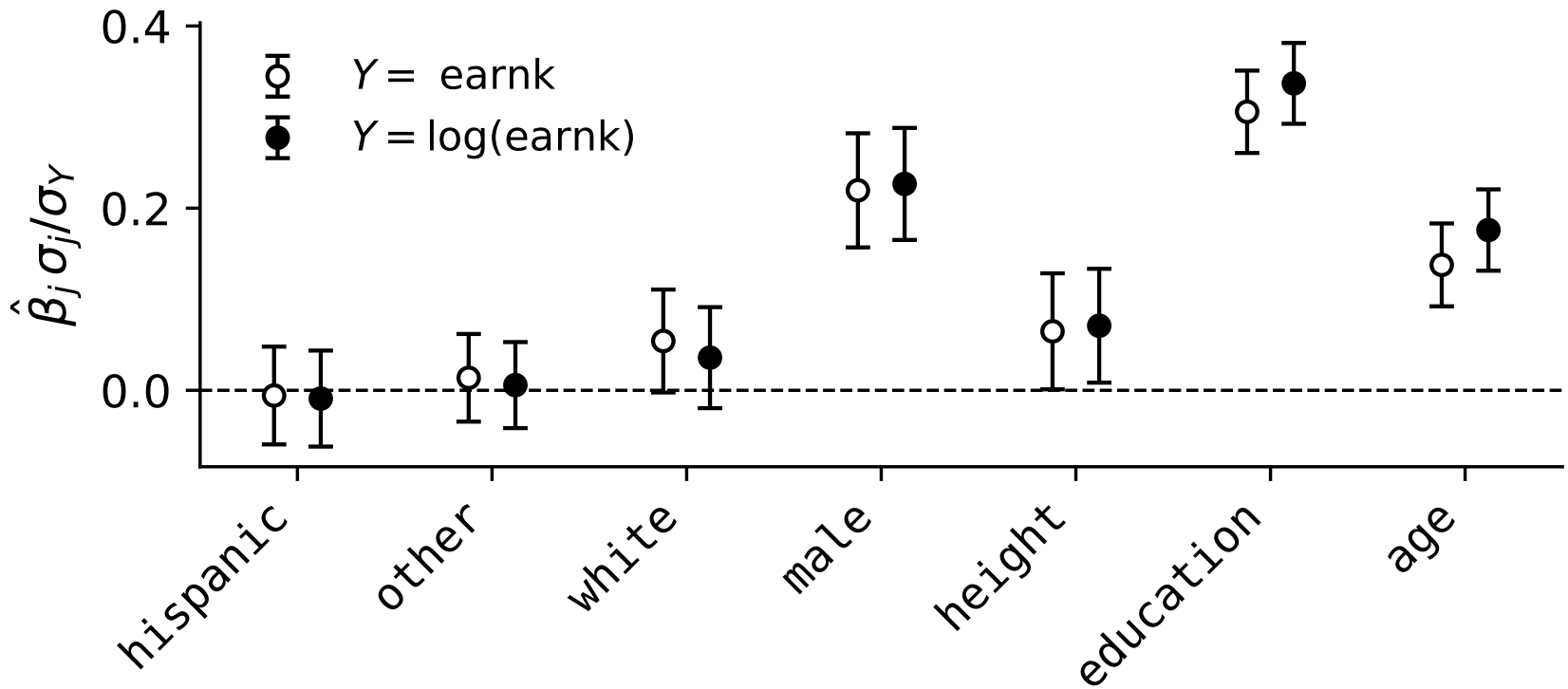
Coefficients vs. scaled coefficients





Standardized coefficients: earnk vs. log(earnk)

The full model fit to the 1627 respondents with positive earnings, with $Y =$ earnk ($R^2 = 0.18$) and with $Y =$ log(earnk) ($R^2 = 0.21$). Each coefficient is scaled by σ_j/σ_Y:



Omitting many predictors

Theorem

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \epsilon$, *where* ϵ *is uncorrelated with* $X_1, \ldots, X_K$.

Omitted variable bias: if $X_2, \ldots, X_K$ *are left out, the slope of the regression of* Y *on* X_1 *is*

$$\boxed{\beta_1' = \beta_1 + \sum_{j=2}^{K} \beta_j \beta_{1,j}'}, \qquad \beta_{1,j}' = \frac{cov(X_1, X_j)}{var(X_1)}.$$

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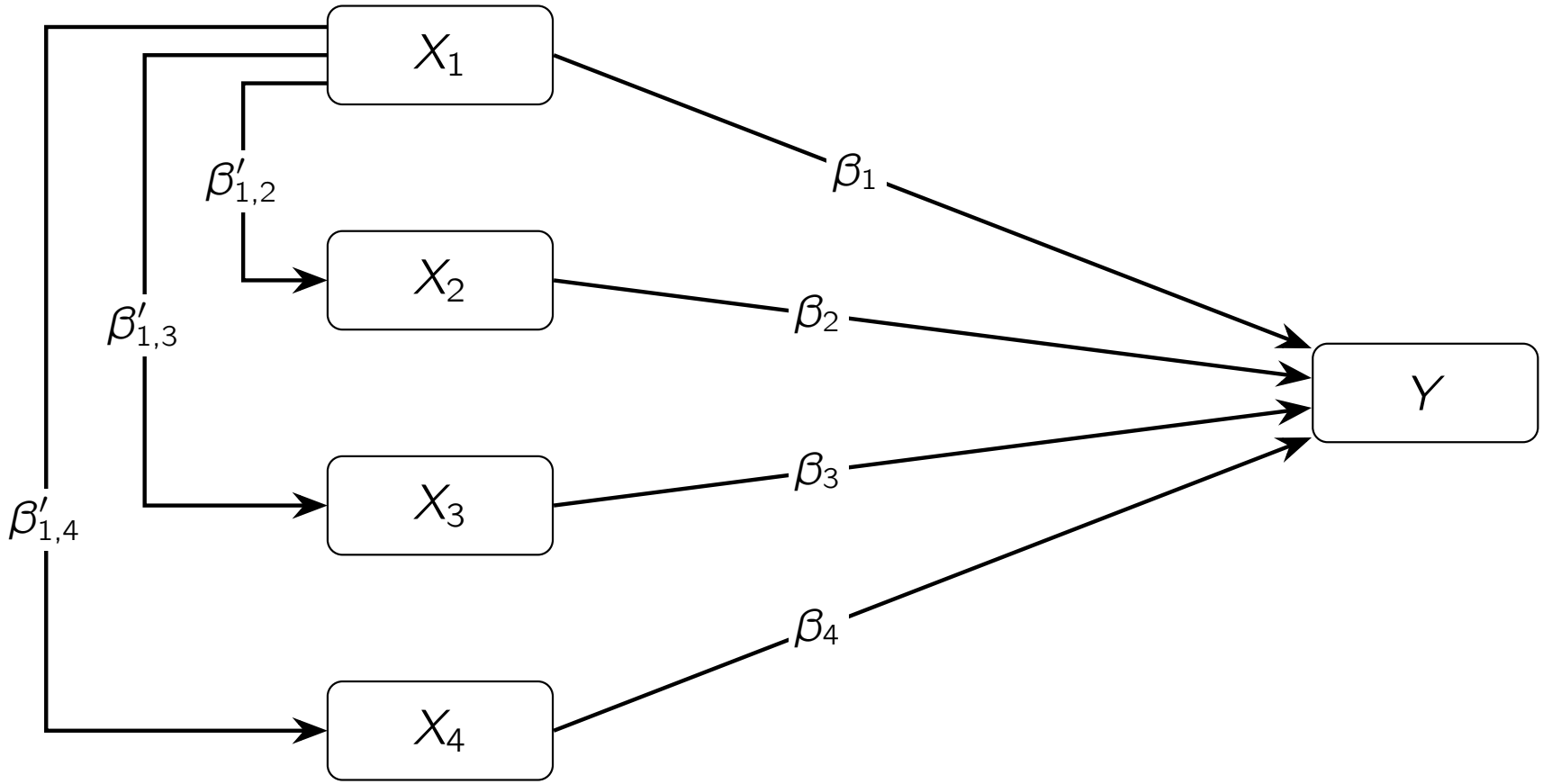
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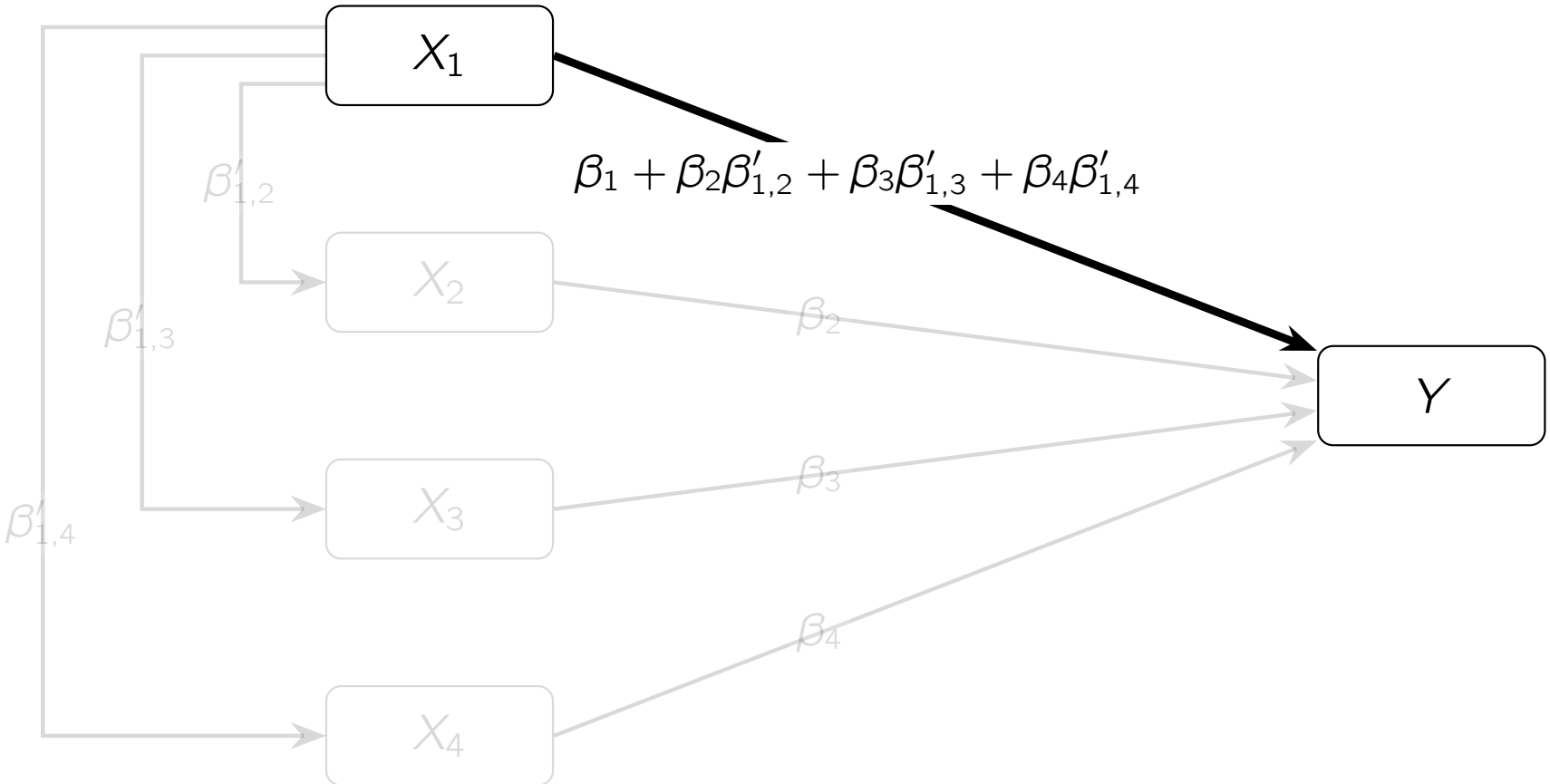
Question

Derive this formula, assuming $E[X_j] = 0$ for $j = 1, \ldots, K$ and $\beta_0 = 0$ (hence $E[Y] = 0$).

Leaving out all predictors but one

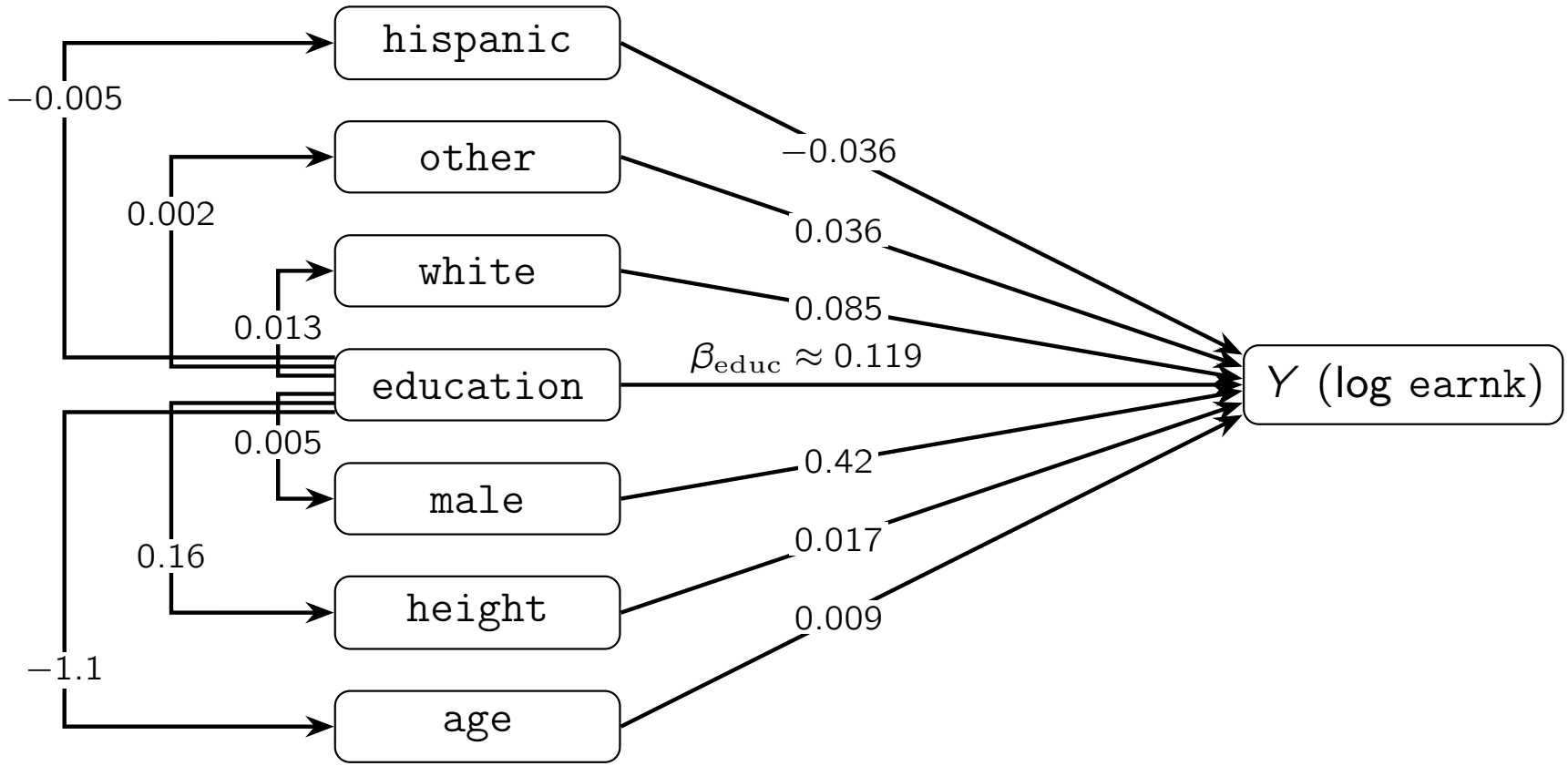


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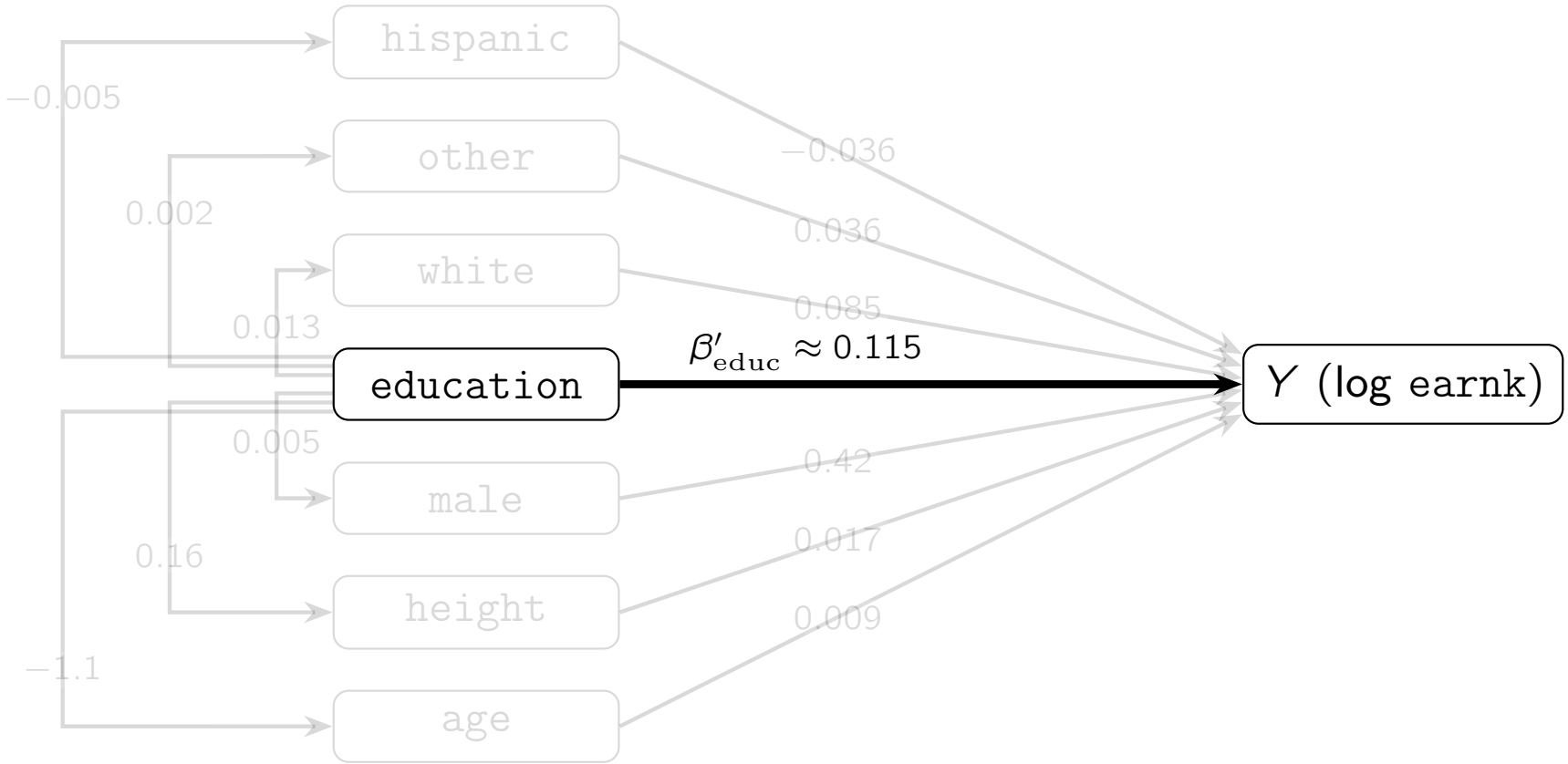
Here $\beta'_{1,j}$ is the single-predictor slope of X_j on X_1: this is the theorem for K predictors.

Leaving out all predictors but education



Left arrows: single-predictor slopes $\beta'_{\text{educ},j}$ of each predictor on `education`. Right arrows: coefficients β_j in the full model.

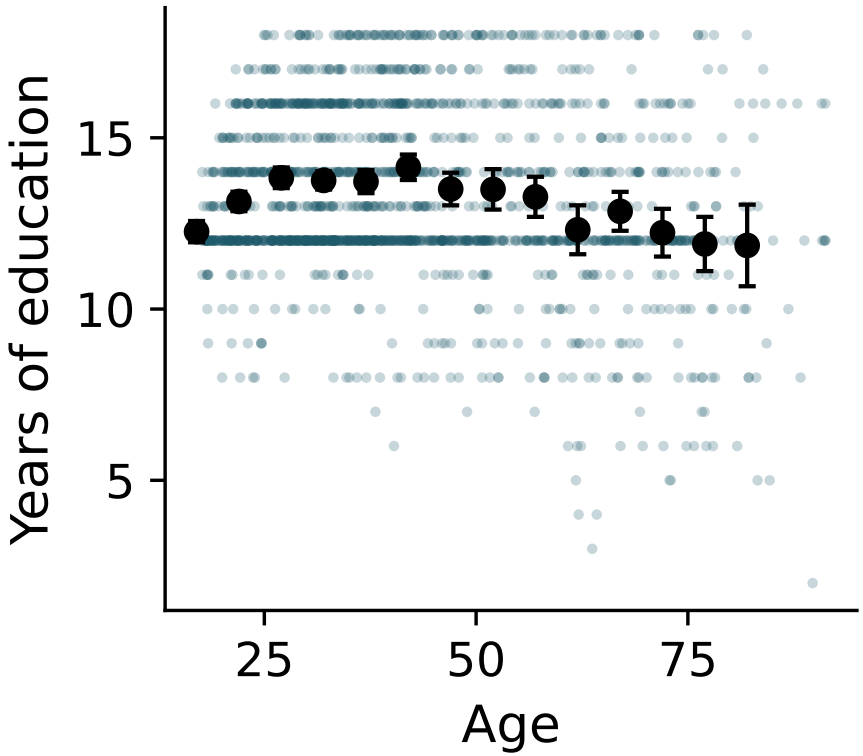
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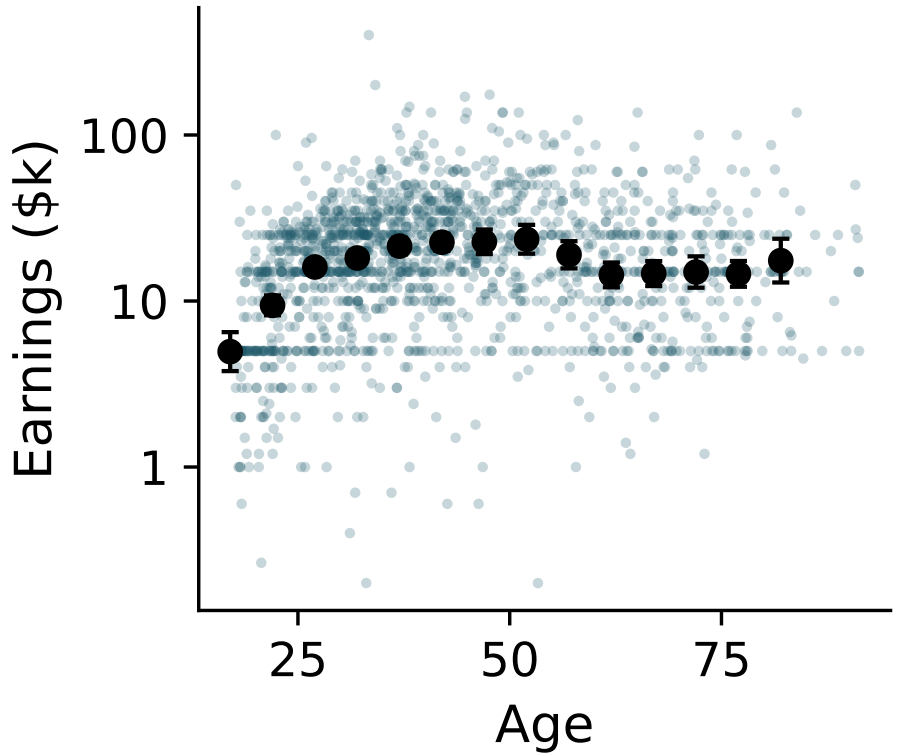


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$$\underbrace{0.115}_{\beta'_{\text{educ}}} \approx \underbrace{0.119}_{\beta_{\text{educ}}} + \underbrace{0.001}_{\text{ethnicity (3 dummies)}} + \underbrace{0.42 \times 0.005}_{\text{male} \approx 0.002} + \underbrace{0.017 \times 0.16}_{\text{height} \approx 0.003} + \underbrace{0.009 \times (-1.1)}_{\text{age} \approx -0.010}$$

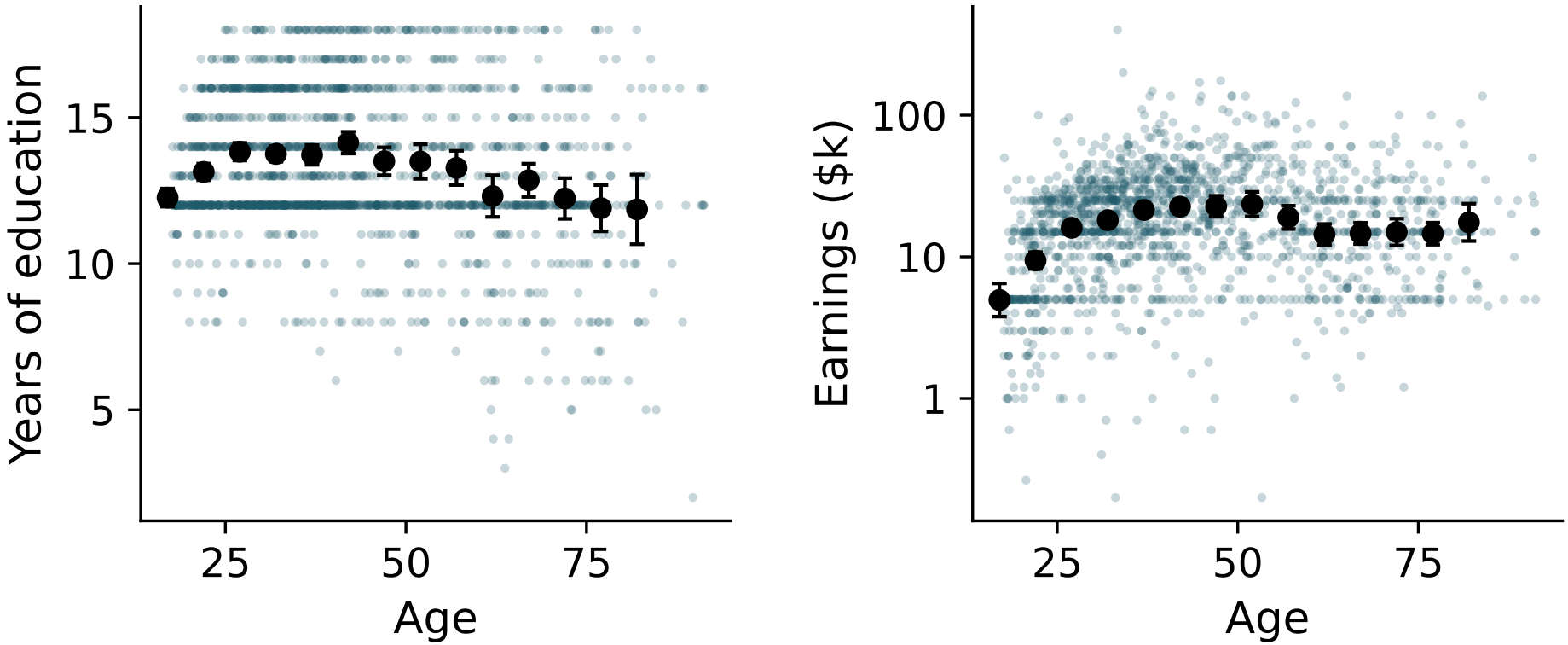
Education and earnings vs. age





Respondents with positive earnings; earnings on a log scale. Black dots: 5-year averages with 95% CIs.

Education and earnings vs. age

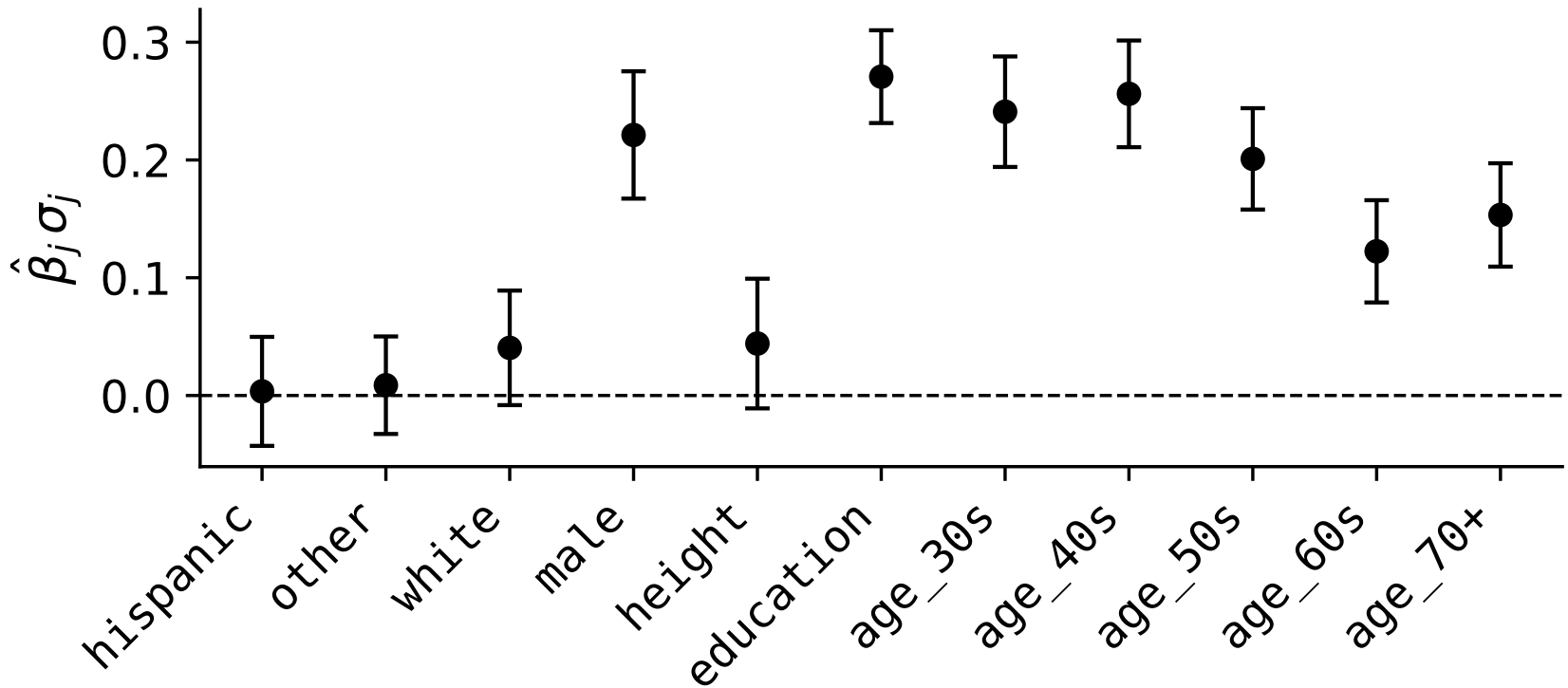


Respondents with positive earnings; earnings on a log scale. Black dots: 5-year averages with 95% CIs.

Explanation: Older respondents are from earlier generations, when less schooling was typical (the youngest are partly still in school).

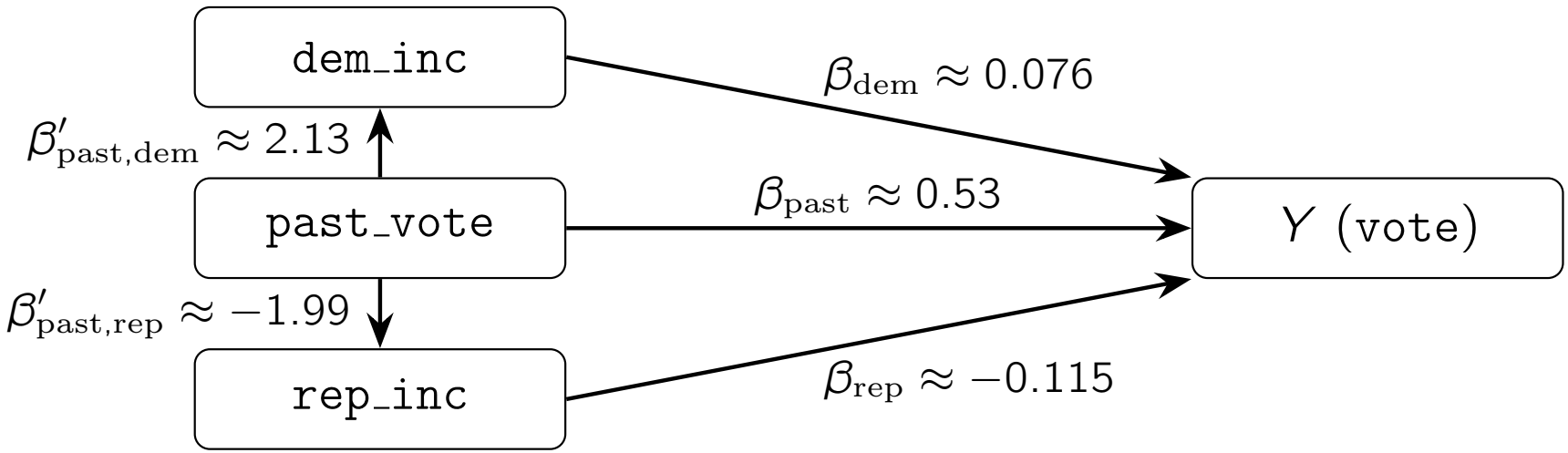
Age as a categorical predictor

Replace age by one dummy variable per decade (baseline: 18–29, with 70 and over pooled), keeping the other predictors:



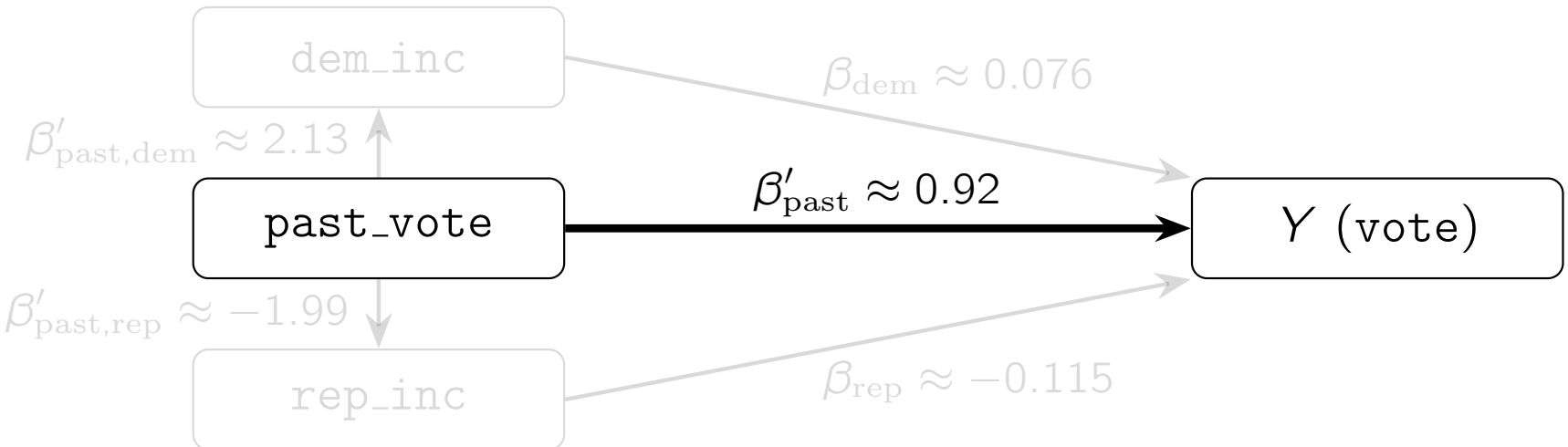
Revisiting elections: comparison to past vote only

Dep. Variable:		vote	R-squared:		0.834
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0382	0.012	3.311	0.001	0.016	0.061
past_vote	0.9182	0.020	46.584	0.000	0.879	0.957



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const	0.0382	0.012	3.311	0.001	0.016	0.061
past_vote	0.9182	0.020	46.584	0.000	0.879	0.957



$$\underbrace{0.92}_{\beta'_{\text{past}}} \approx \underbrace{0.53}_{\beta_{\text{past}}} + \underbrace{0.076 \times 2.13}_{\beta_{\text{dem}}\beta'_{\text{past,dem}} \approx 0.16} + \underbrace{(-0.115) \times (-1.99)}_{\beta_{\text{rep}}\beta'_{\text{past,rep}} \approx 0.23}$$

Districts with a high past vote tend to have Democratic incumbents, and incumbency raises the vote.

Revisiting children's test scores: mother age

Children's test scores again, now adding X_{age} (mom_age), the mother's age at birth.

Dep. Variable:		kid_score	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	70.9569	8.307	8.542	0.000	54.631	87.283
mom_age	0.6952	0.362	1.920	0.055	-0.016	1.407

Dep. Variable:		kid_score	R-squared:			0.215
	coef	std err	t	P>|t|	[0.025	0.975]
const	20.9847	9.130	2.298	0.022	3.039	38.930
mom_hs	5.6472	2.258	2.501	0.013	1.210	10.085
mom_iq	0.5625	0.061	9.276	0.000	0.443	0.682
mom_age	0.2248	0.331	0.680	0.497	-0.425	0.875

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Question

Interpret the coefficient on `mom_age` in each model. With only `mom_hs` and `mom_iq`, the coefficients were 5.95 and 0.564 and $R^2 = 0.214$. Why did adding `mom_age` change so little even though age is associated with test scores in the first regression? What does this tell us about association between age and the other predictors?

Mother's age vs. schooling and IQ

Dep. Variable:		mom_hs	R-squared:			0.046
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0423	0.164	0.258	0.797	-0.280	0.365
mom_age	0.0326	0.007	4.565	0.000	0.019	0.047
Dep. Variable:		mom_iq	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	88.4081	6.105	14.482	0.000	76.409	100.407
mom_age	0.5087	0.266	1.912	0.057	-0.014	1.032

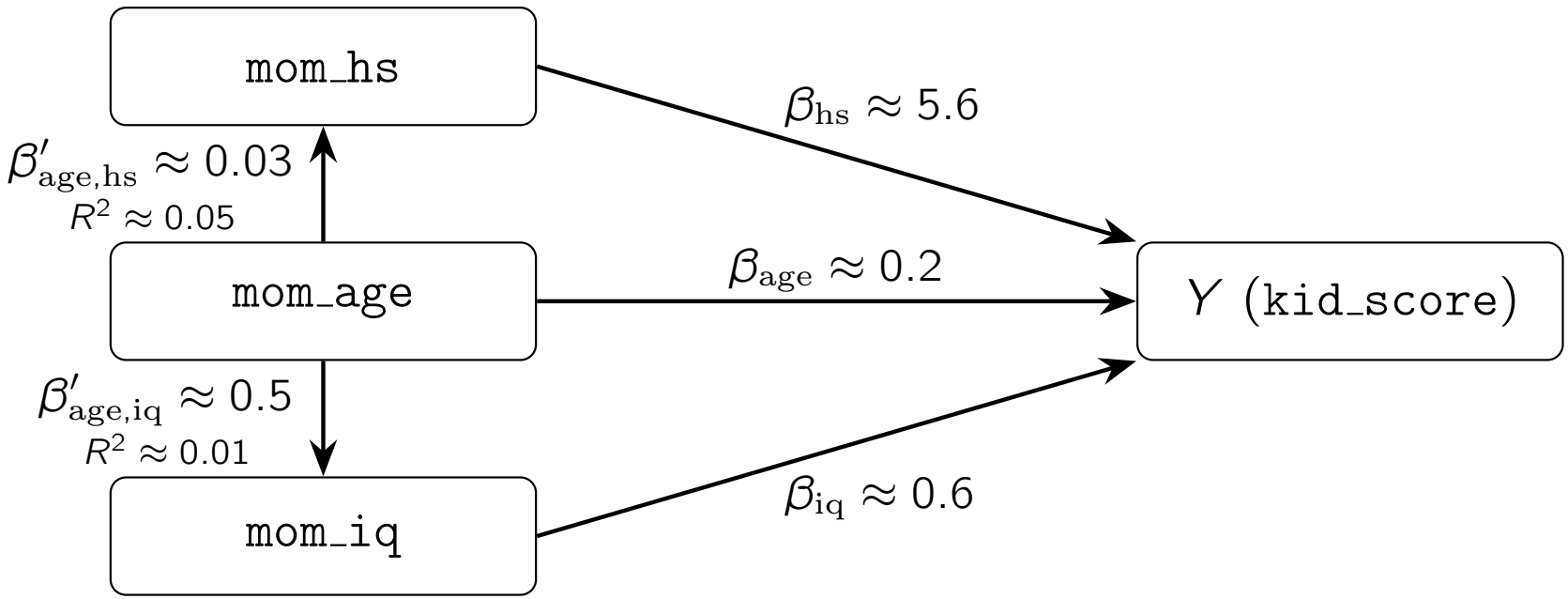
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```
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                 coef    std err          t      P>|t|      [0.025      0.975]
const          0.0423      0.164      0.258      0.797      -0.280       0.365
mom_age        0.0326      0.007      4.565      0.000       0.019       0.047
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const         88.4081      6.105     14.482      0.000      76.409     100.407
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```

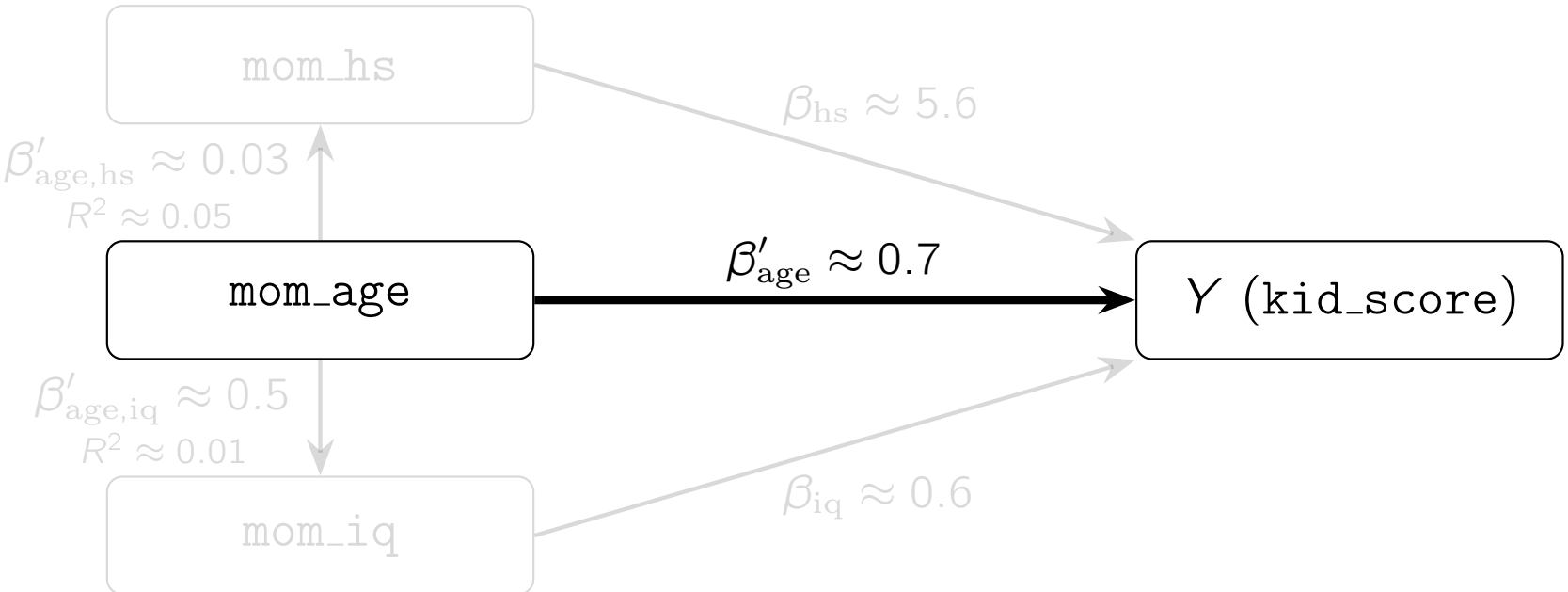
Question

Use the omitted variable bias formula to write the slope $\beta'_{\text{age}} \approx 0.70$ of the single-predictor regression of `kid_score` on `mom_age` in terms of the coefficients of the three-predictor model and the slopes $\beta'_{\text{age,hs}}$ and $\beta'_{\text{age,iq}}$ above.

How mother's age is related to test scores



How mother's age is related to test scores



$$\underbrace{0.7}_{\beta'_{\text{age}}} \approx \underbrace{0.2}_{\beta_{\text{age}}} + \underbrace{0.03 \times 5.6}_{\beta'_{\text{age,hs}}\beta_{\text{hs}} \,\approx\, 0.2} + \underbrace{0.5 \times 0.6}_{\beta'_{\text{age,iq}}\beta_{\text{iq}} \,=\, 0.3}$$

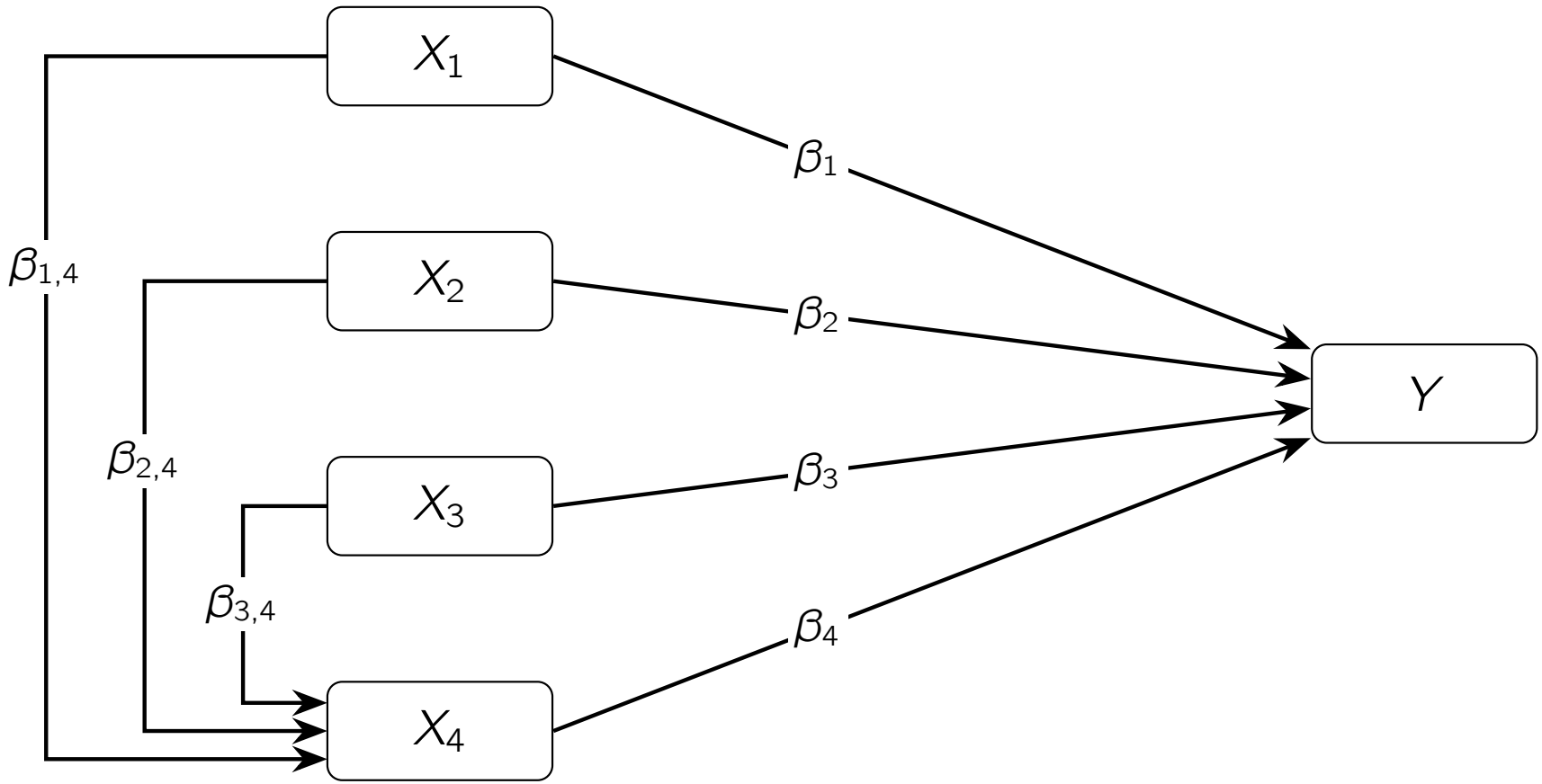
Omitting one predictor

Theorem

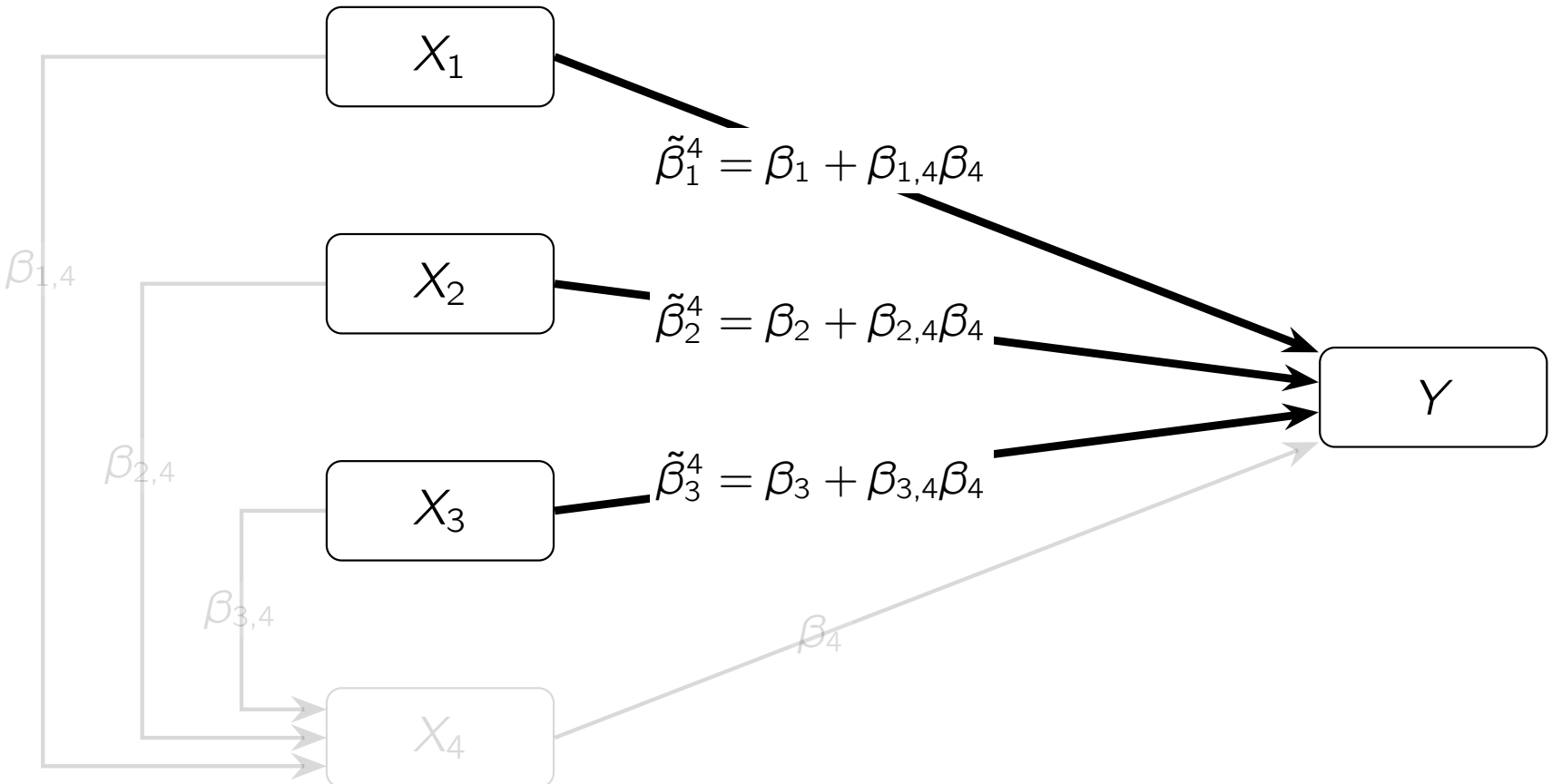
Suppose $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_K X_K + \epsilon$, *where* ϵ *is uncorrelated with* $X_1, \ldots, X_K$.

> *Omitted variable bias: if only* X_K *is left out, the coefficient on* X_j $(j < K)$ *in the regression of* Y *on* $X_1, \ldots, X_{K-1}$ *is*
>
> $$\boxed{\tilde{\beta}_j^K = \beta_j + \beta_K \beta_{j,K}},$$
>
> *where* $\beta_{j,K}$ *is the coefficient on* X_j *in the regression of* X_K *on* $X_1, \ldots, X_{K-1}$.

Leaving out one predictor of many

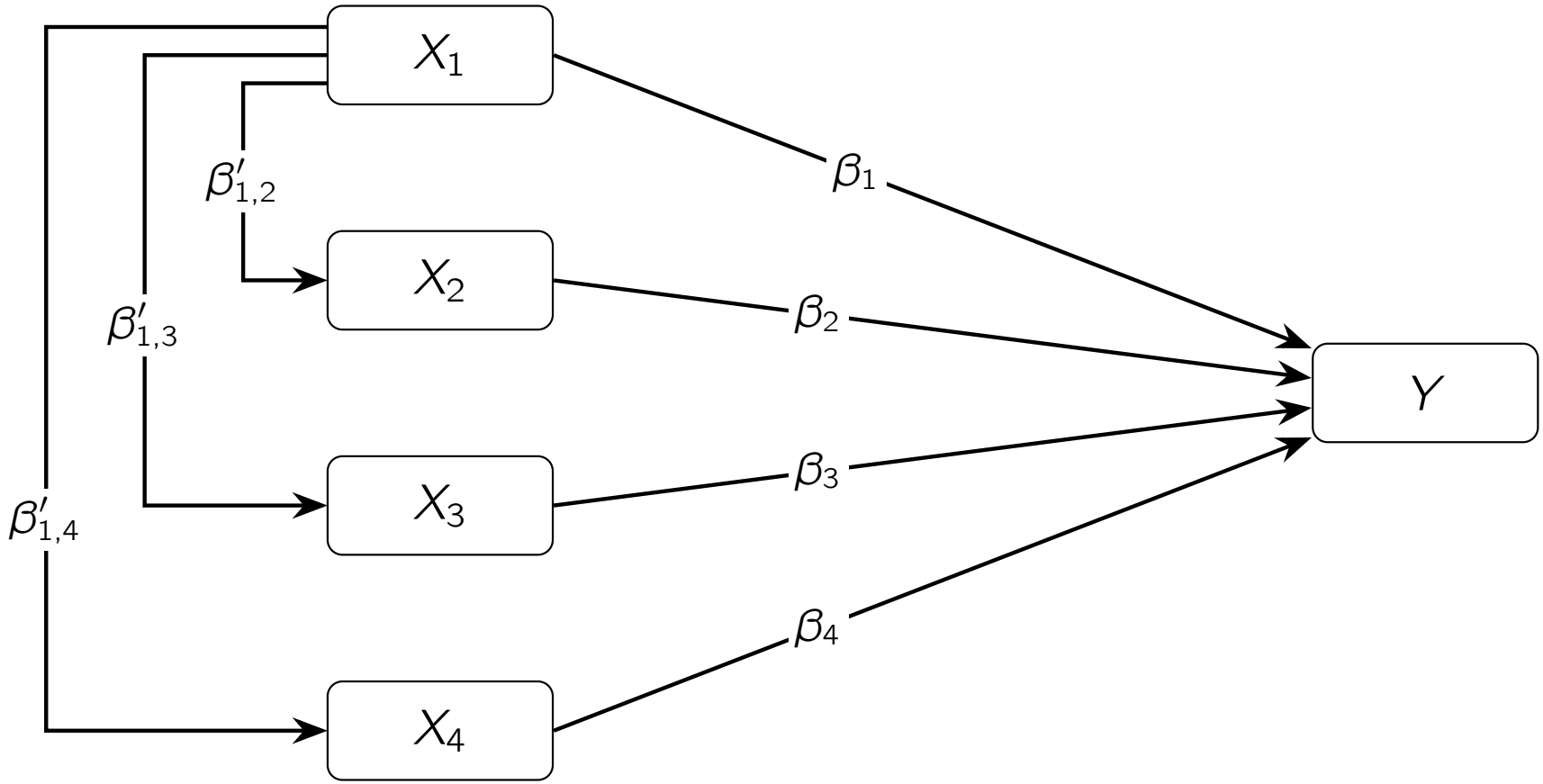


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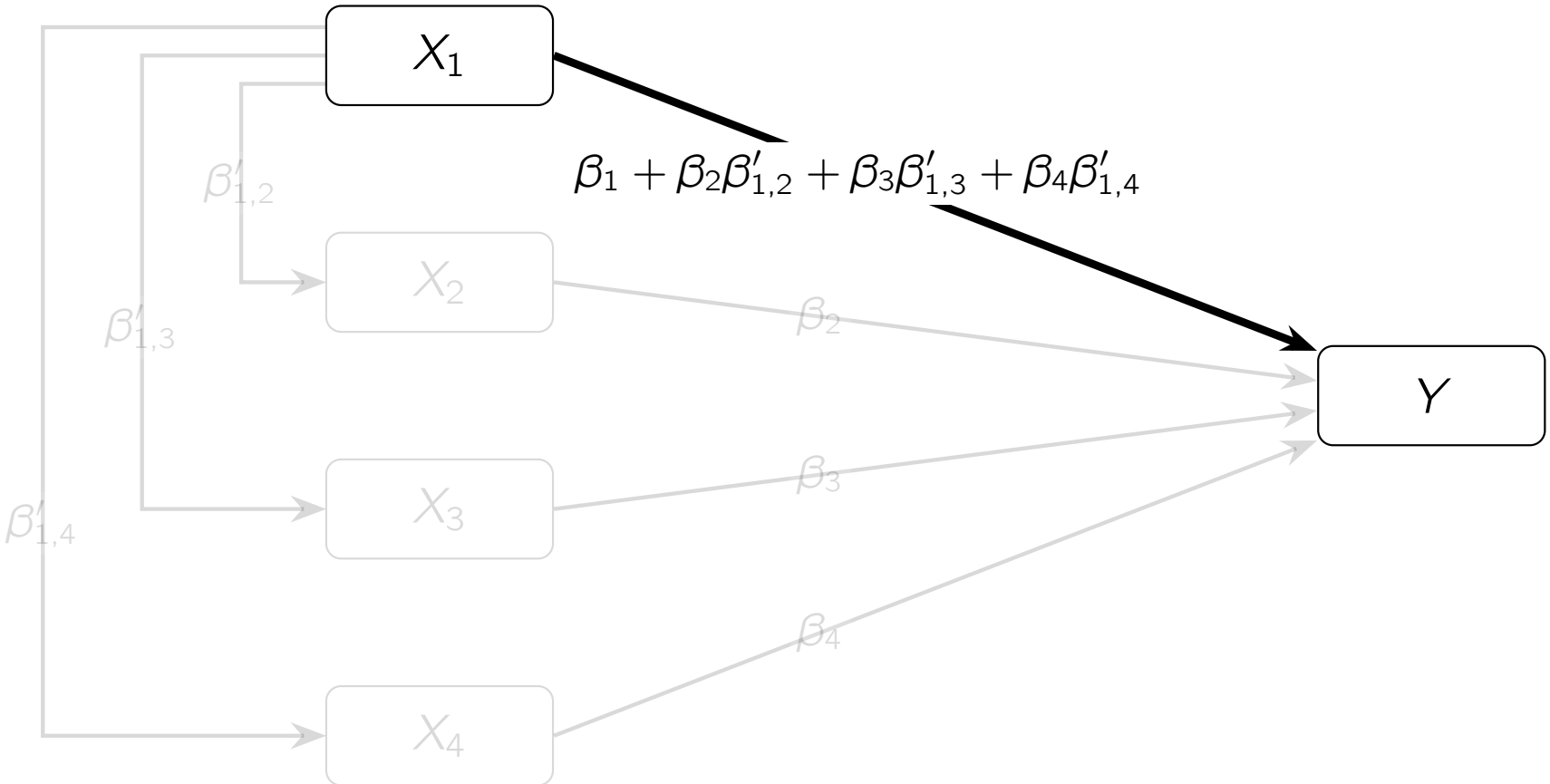


Each remaining coefficient picks up β_4 times the coefficient of its predictor in the regression of X_4 on X_1, X_2, X_3.

Compared to OVB Theorem from earlier

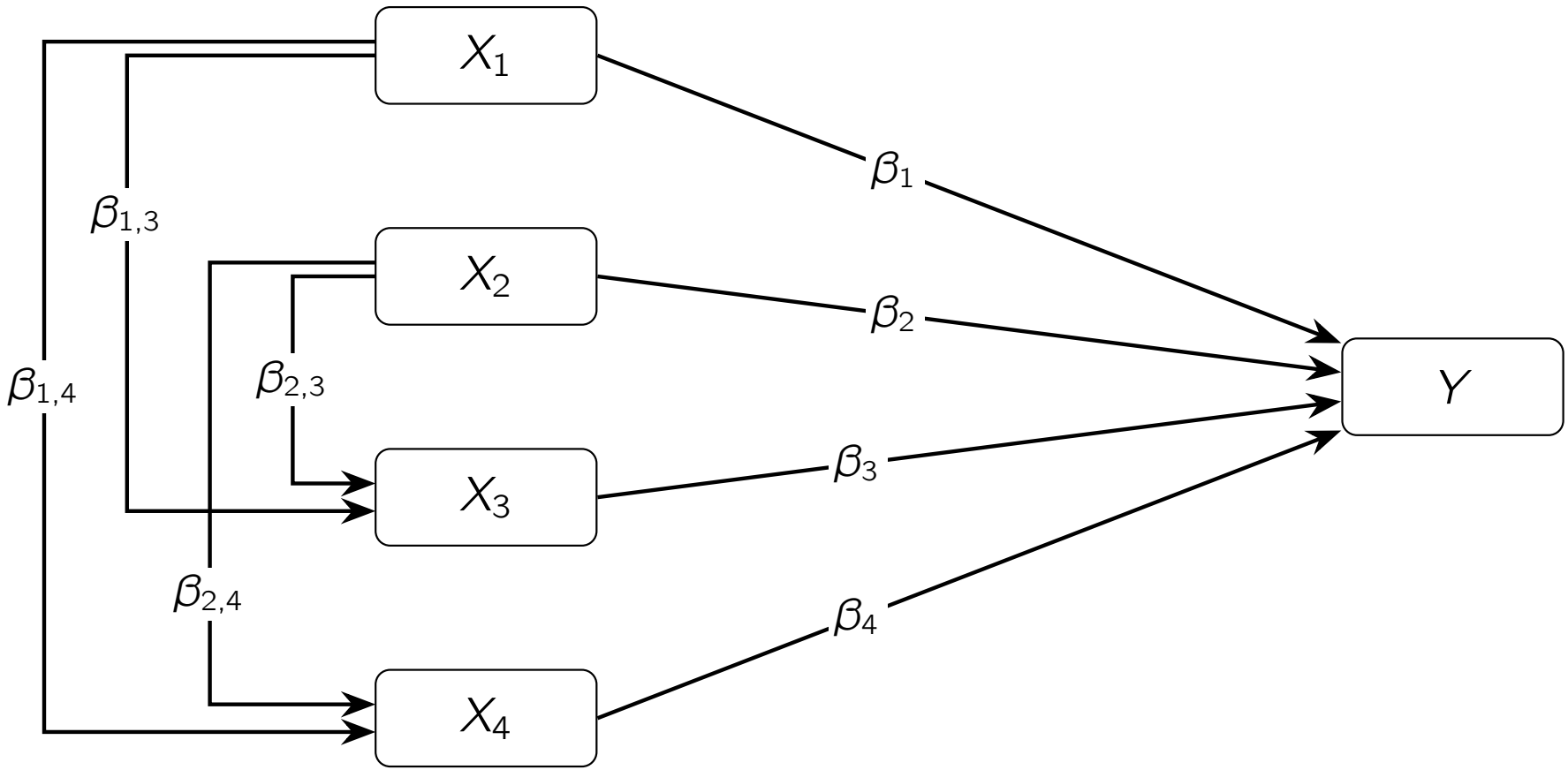


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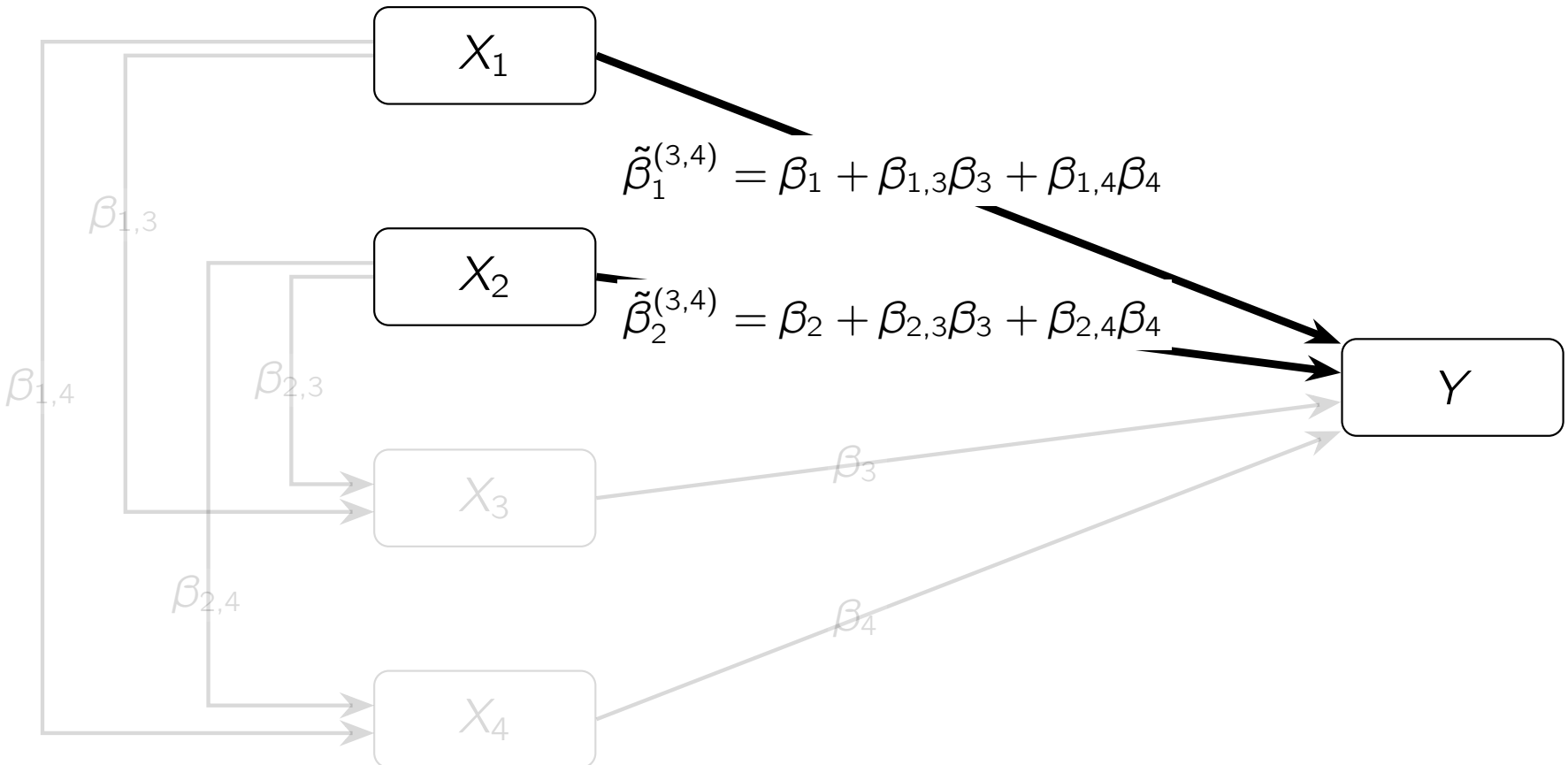
Here $\beta'_{1,j}$ is the single-predictor slope of X_j on X_1: this is the theorem for K predictors.

Leaving out any subset of predictors



$\beta_{j,k}$ is the coefficient on X_j in the regression of the left-out X_k on the kept predictors X_1, X_2. With one kept predictor these are the single-predictor slopes $\beta'_{1,k}$; with one left-out predictor this is the formula for omitting one predictor.

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Earnings without ethnicity

Leave out only the ethnicity dummies:

Dep. Variable:		log_earnk	R-squared:		0.209
	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.5728	0.488	-1.173	0.241	-1.531	0.385
male	0.4095	0.057	7.128	0.000	0.297	0.522
height	0.0180	0.007	2.445	0.015	0.004	0.033
education	0.1204	0.008	15.171	0.000	0.105	0.136
age	0.0094	0.001	8.022	0.000	0.007	0.012

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age	0.0094	0.001	8.022	0.000	0.007	0.012

Each dummy adds a term $\beta_r \beta_{j,r}$, where $\beta_{j,r}$ is the coefficient on X_j in the regression of the dummy X_r on `male`, `height`, `education`, `age` (terms in the order hispanic, other, white):

$$\begin{aligned}\tilde{\beta}_{\text{male}}^{(\text{hispanic, other, white})} &\approx 0.418 + (-0.036)(0.061) + (0.036)(0.024) \\ &\quad + (0.085)(-0.079) \approx 0.410\end{aligned}$$

$$\begin{aligned}\tilde{\beta}_{\text{educ}}^{(\text{hispanic, other, white})} &\approx 0.119 + (-0.036)(-0.006) + (0.036)(0.002) \\ &\quad + (0.085)(0.014) \approx 0.120\end{aligned}$$

Menarche and menopause

"It remains uncertain whether age at menarche is associated with age at menopause. Some studies report that women with early menarche also have early menopause. Other studies report that women with early menarche have late menopause, or they report no association. The duration of the reproductive period may be an indicator of the cumulative endogenous exposure to estrogens and progestogens during life course and is associated with risk of breast cancer and endometrial cancer."

Bjelland et al., *Human Reproduction* (2018)

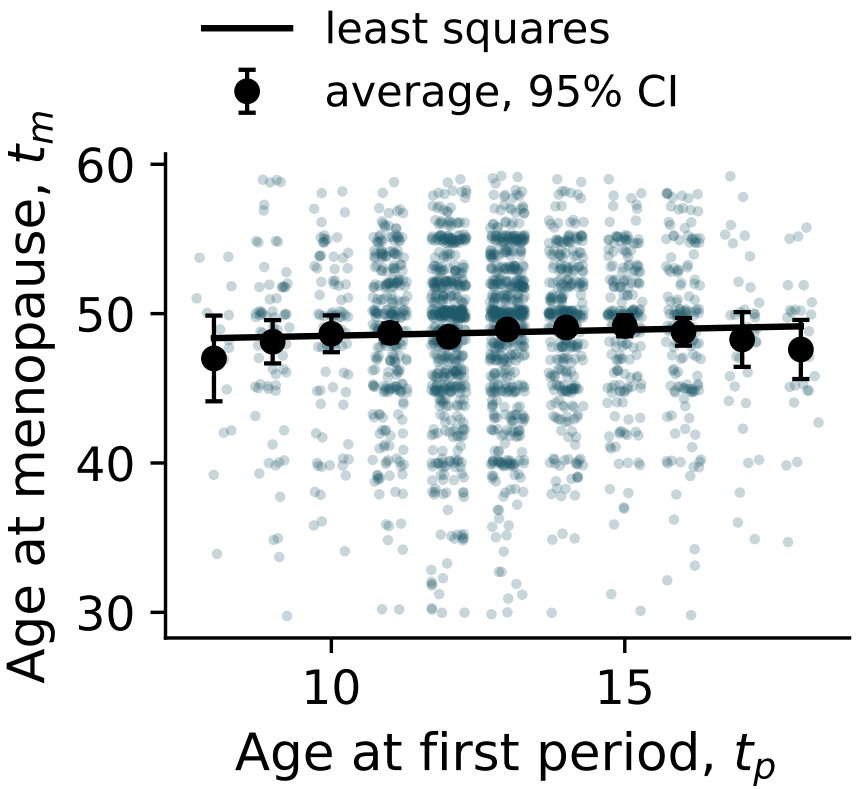
Example

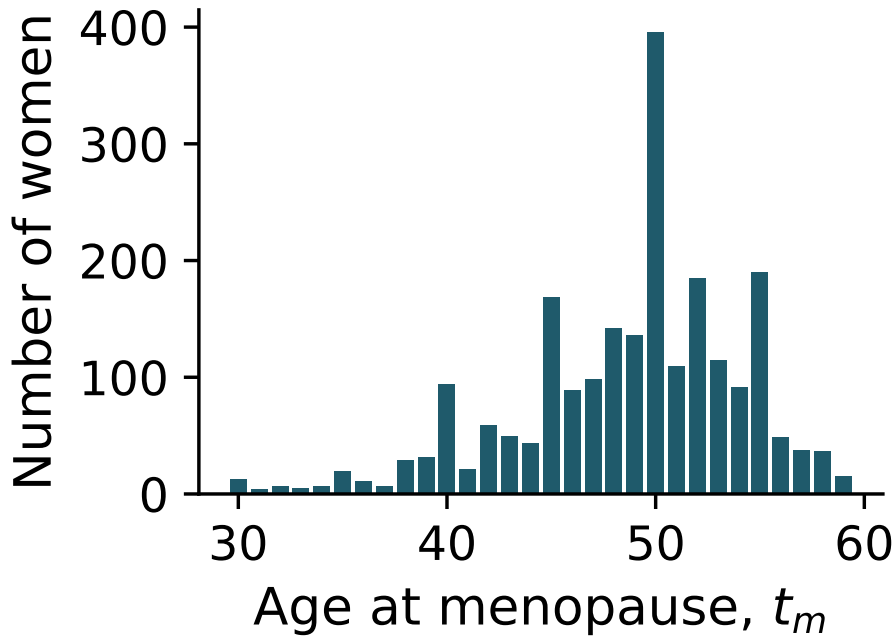
What factors influence the timing of menopause? Data from NHANES 2013–2018 survey of $N = 2248$ US women.

- t_m (`tm`): age at menopause (last period)
- t_p (`tp`): age at first period (puberty)
- X_{smoker} (`smoker`): 1 if she has smoked at least 100 cigarettes
- X_{births} (`births`): number of live births
- $X_{\text{birthyear}}$ (`birthyear`): year of birth

All ages are recalled at the interview.

t_m **vs.** t_p


least squares
average, 95% CI
Age at menopause, t_m
60
50
40
30
10
15
Age at first period, t_p


Number of women
400
300
200
100
0
30
40
50
60
Age at menopause, t_m

Dep. Variable:			tm	R-squared:		0.001
	coef	std err	t	P>|t|	[0.025	0.975]
const	47.7292	0.805	59.256	0.000	46.150	49.309
tp	0.0790	0.062	1.276	0.202	-0.042	0.200

Dep. Variable:			tm	R-squared:		0.015
	coef	std err	t	P>|t|	[0.025	0.975]
const	165.0489	24.430	6.756	0.000	117.140	212.958
tp	0.0522	0.063	0.827	0.409	-0.072	0.176
smoker	-0.6673	0.239	-2.798	0.005	-1.135	-0.200
births	-0.1288	0.062	-2.066	0.039	-0.251	-0.007
birthyear	-0.0596	0.012	-4.778	0.000	-0.084	-0.035

The second model uses the 2168 women with smoking and births recorded (the first uses all 2248).

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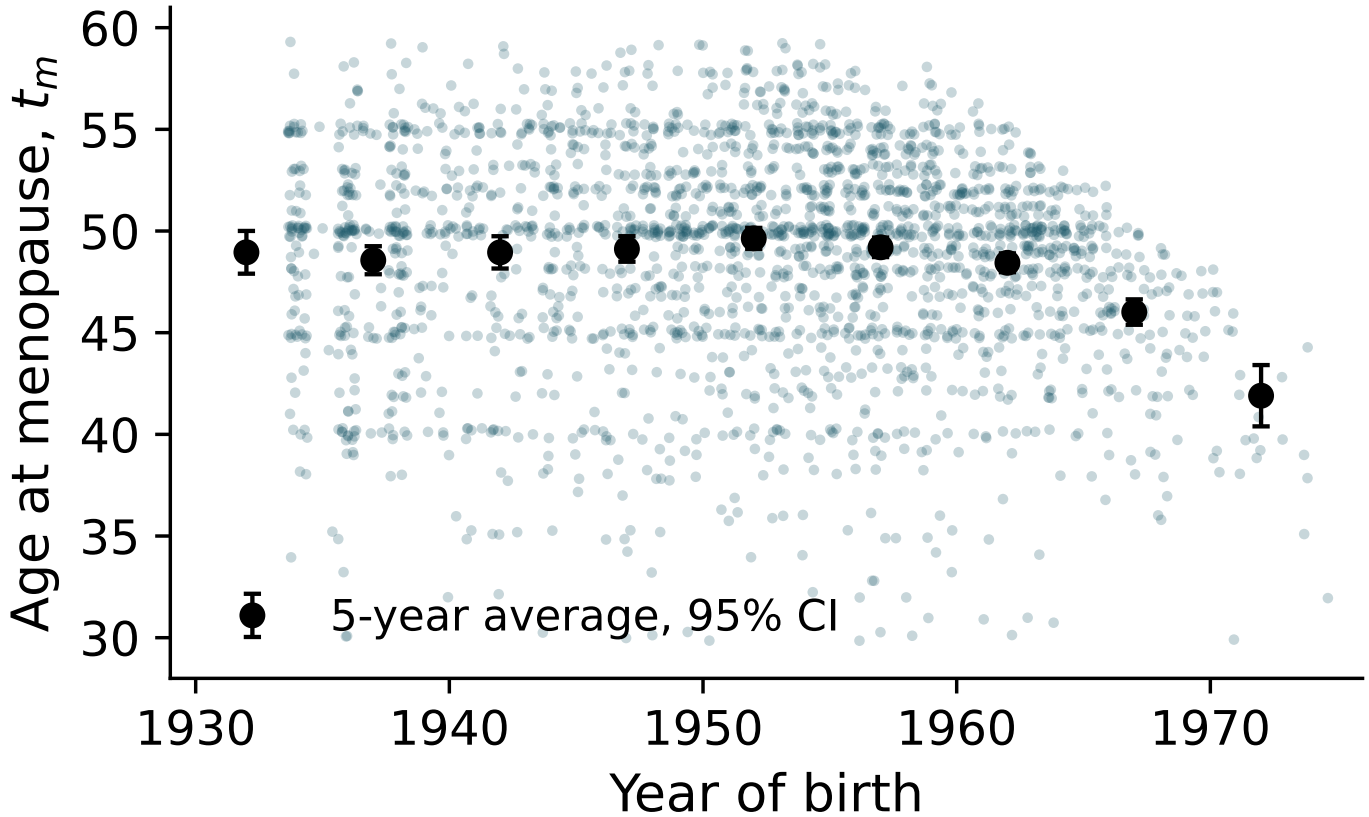
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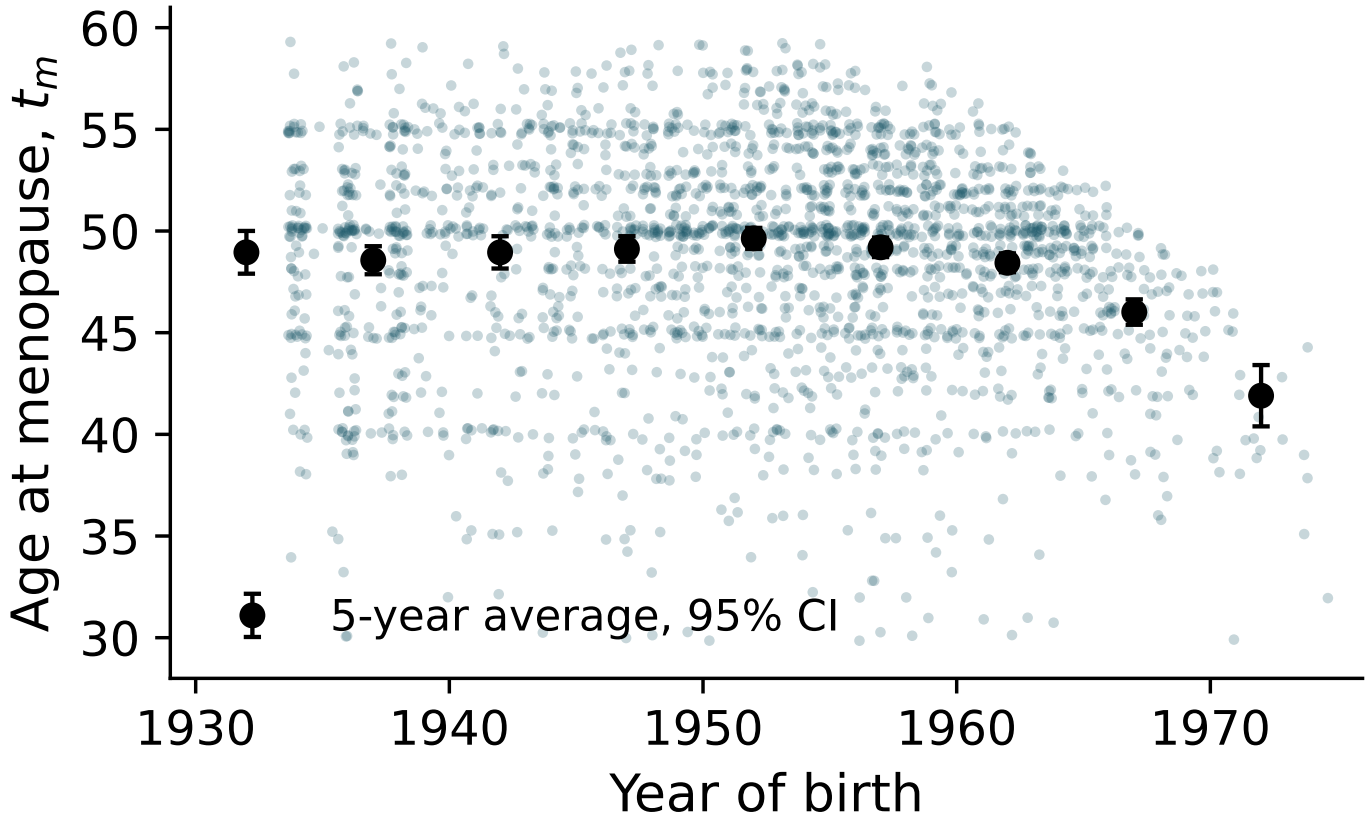
Question

Interpret the coefficients. `smoker` and `birthyear` are clearly significant, yet $R^2 = 0.015$. What does that tell you?

t_m vs. year of birth

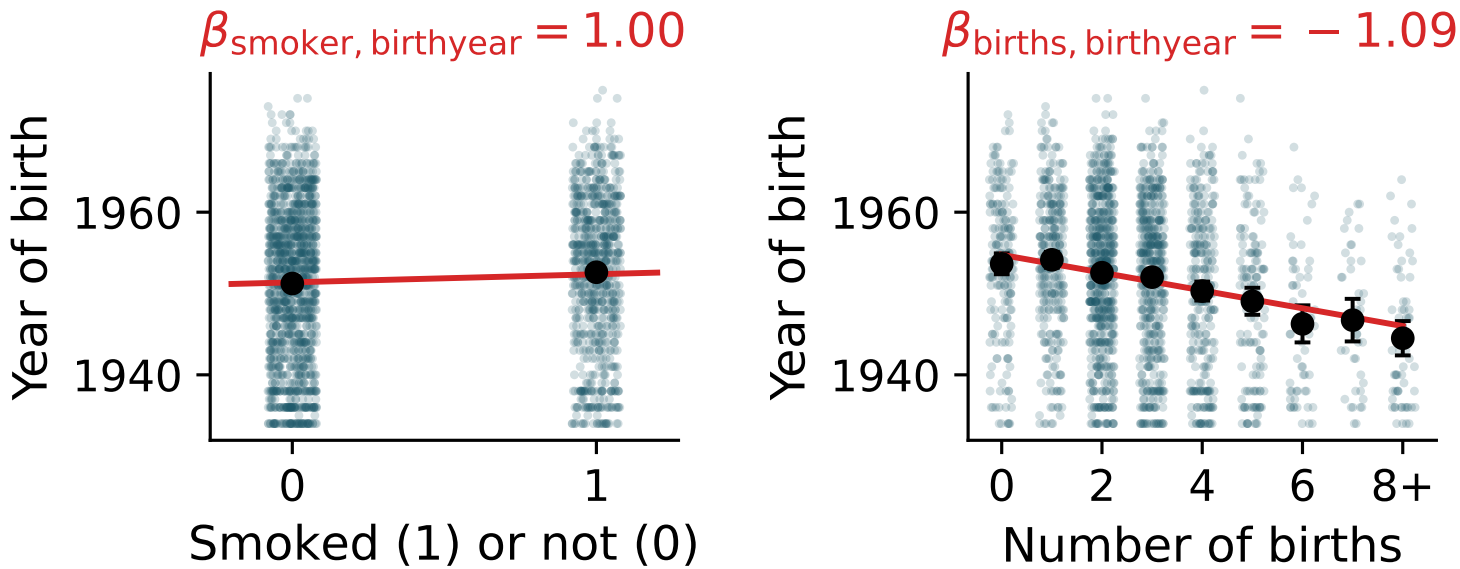


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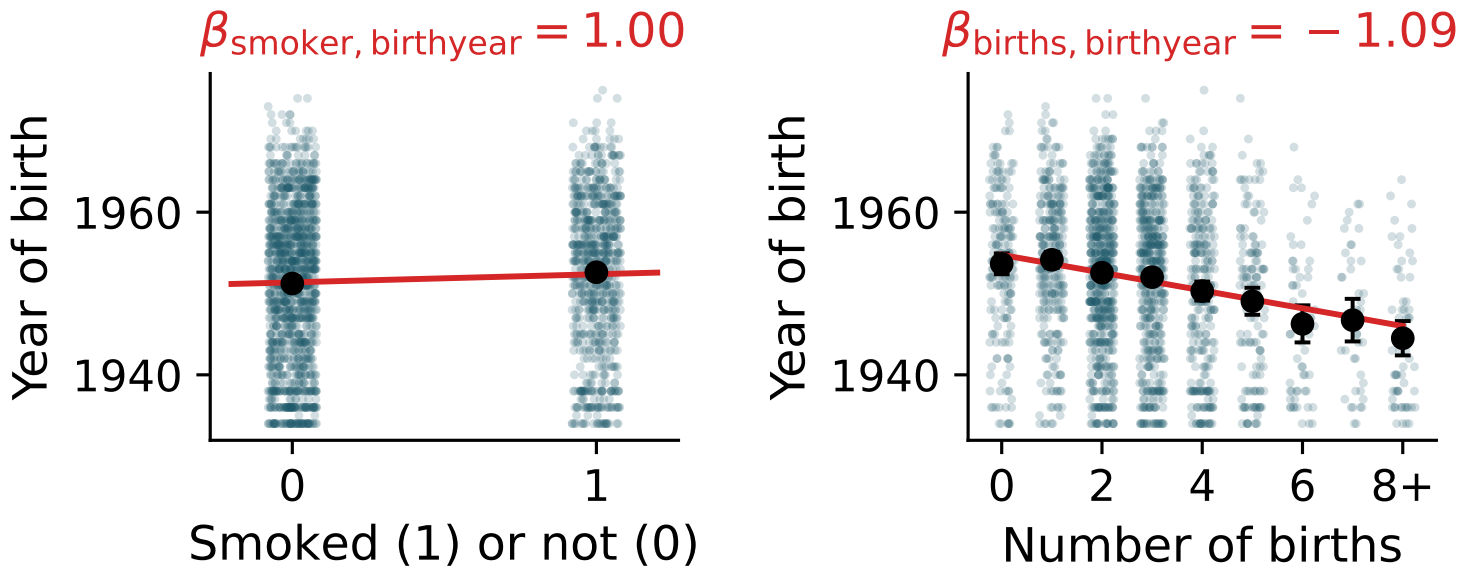
This relation, which is the most significant result, is actually an artifact of the way the data was collected (the dataset includes only women who have gone through menopause!)

Year of birth vs. smoking and births



Dep. Variable:		birthyear	R-squared:			0.054
	coef	std err	t	P>|t|	[0.025	0.975]
const	1954.4093	0.389	5022.637	0.000	1953.646	1955.172
smoker	1.0310	0.410	2.516	0.012	0.227	1.835
births	-1.1077	0.104	-10.600	0.000	-1.313	-0.903

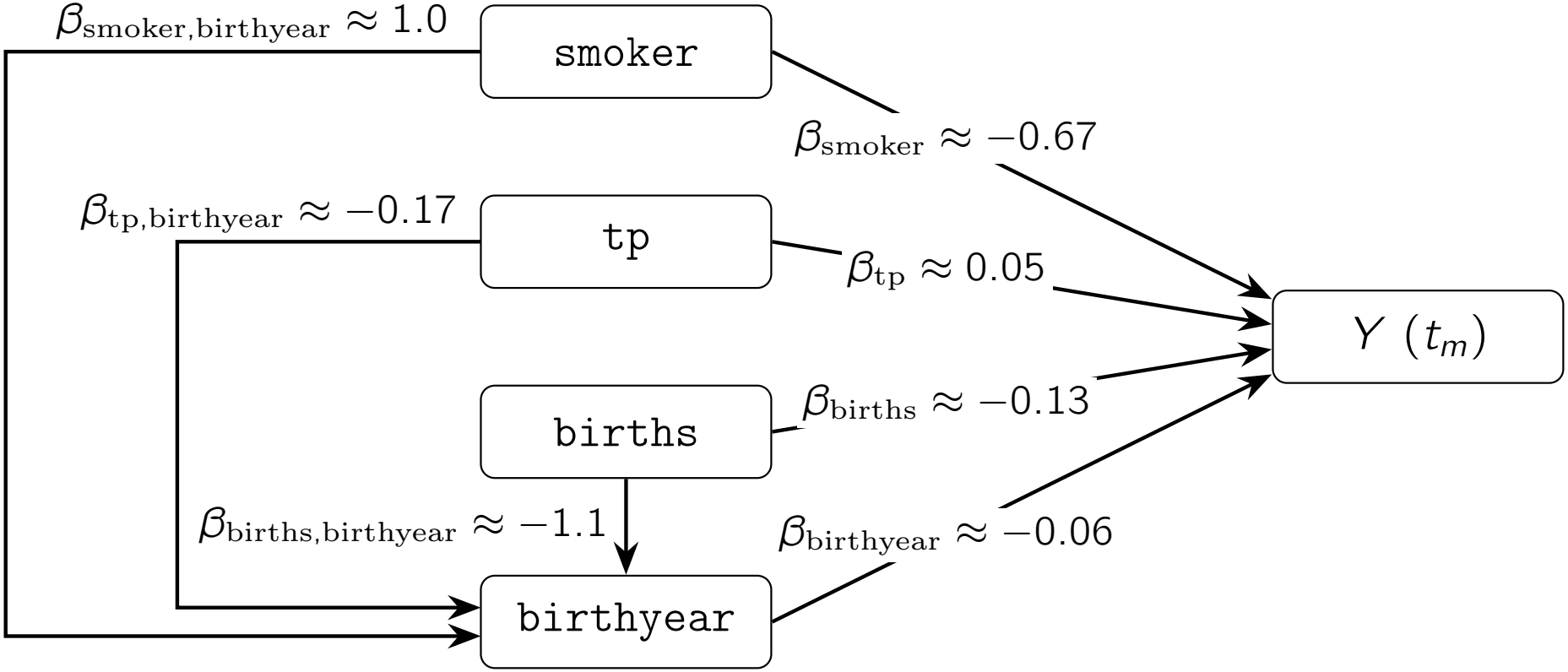
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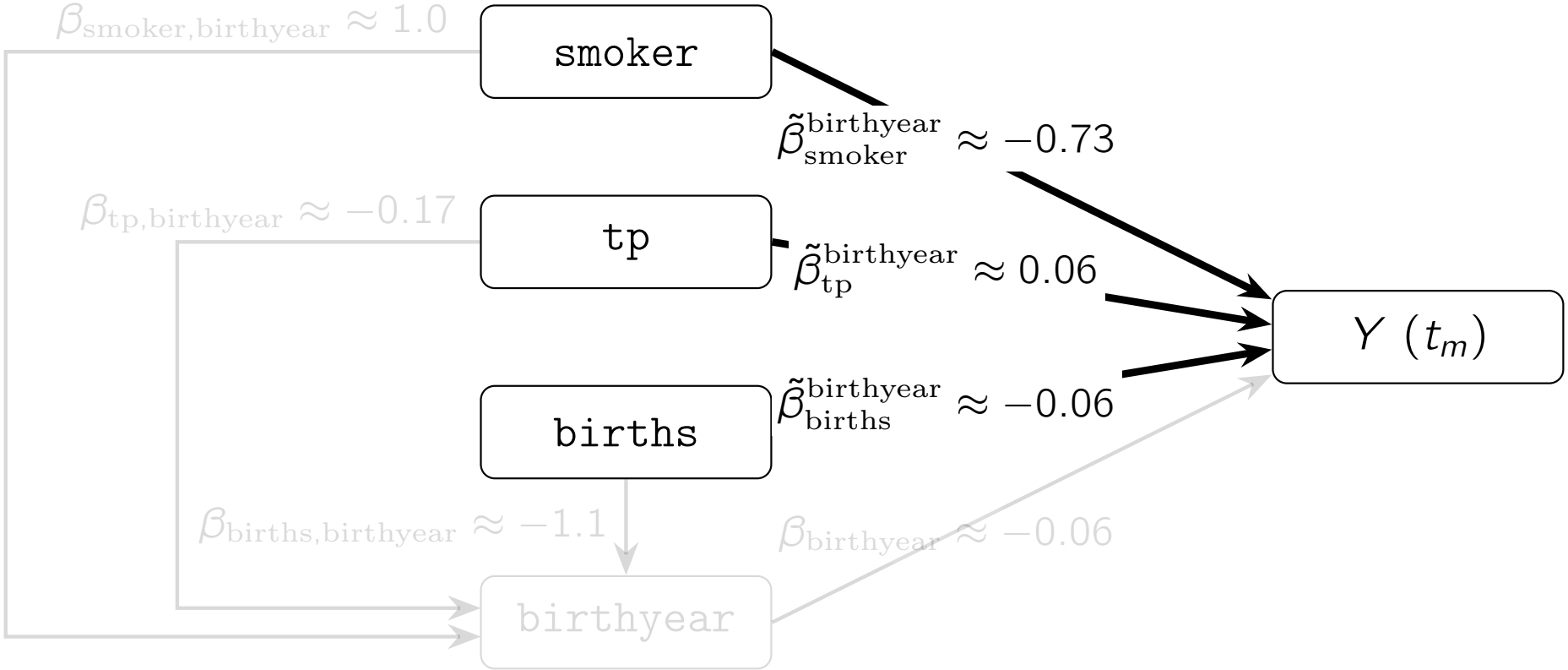
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births	-1.1077	0.104	-10.600	0.000	-1.313	-0.903

Question

Without refitting the model, can you predict how these trends affect the coefficients on `smoker` and `births` when `birthyear` is left out of the model? (do they go up or down?)



How do the coefficients on `smoker` and `births` change if `birthyear` is left out?



How do the coefficients on `smoker` and `births` change if `birthyear` is left out?

$$\underbrace{-0.73}_{\tilde{\beta}^{\text{birthyear}}_{\text{smoker}}} \approx \underbrace{-0.67}_{\beta_{\text{smoker}}} + \underbrace{1.0 \times (-0.06)}_{\beta_{\text{smoker,birthyear}}\beta_{\text{birthyear}} = -0.06}$$

$$\underbrace{-0.06}_{\tilde{\beta}^{\text{birthyear}}_{\text{births}}} \approx \underbrace{-0.13}_{\beta_{\text{births}}} + \underbrace{(-1.1) \times (-0.06)}_{\beta_{\text{births,birthyear}}\beta_{\text{birthyear}} \approx 0.07}$$

Age at puberty as a categorical predictor

Replace t_p by one dummy variable per age at first period (baseline: 13, the groups of Bjelland et al.), keeping `smoker`, `births`, `birthyear`:

Dep. Variable:			tm	R-squared:		0.019
	coef	std err	t	P>|t|	[0.025	0.975]
const	167.2685	24.434	6.846	0.000	119.353	215.184
tp_<=9	-1.0247	0.634	-1.617	0.106	-2.267	0.218
tp_10	-0.3945	0.640	-0.617	0.538	-1.649	0.860
tp_11	-0.3522	0.405	-0.869	0.385	-1.147	0.443
tp_12	-0.5932	0.329	-1.803	0.072	-1.238	0.052
tp_14	0.0255	0.383	0.067	0.947	-0.725	0.776
tp_15	0.0200	0.452	0.044	0.965	-0.867	0.907
tp_>=16	-0.6471	0.443	-1.460	0.145	-1.516	0.222
smoker	-0.6574	0.239	-2.748	0.006	-1.126	-0.188
births	-0.1318	0.062	-2.113	0.035	-0.254	-0.009
birthyear	-0.0603	0.013	-4.822	0.000	-0.085	-0.036

Age at puberty as a categorical predictor

Coefficients on the t_p dummies (difference in t_m from $t_p = 13$, 95% CIs):

